

Stat Phys 2 21/01

$N \gg 1$, thermo limit equilibrium: $\left\{ \begin{array}{l} \text{microcanonical} \\ \text{canonical} \\ \text{grand canonical} \end{array} \right.$ ensembles $\begin{array}{l} E = \text{const} \quad N = \text{const} \\ \langle E \rangle \approx E_F \approx \text{const} \end{array}$

Systems: (1) Harmonic Oscillators
- sound wave propagation in a solid

(2) Free system, particles in a "box"
Fermions & bosons

(3) Magnetic Dipoles, localised or not no momentum

(4) Ising Model in 1 dimension

Microcanonical	Canonical	Grand Canonical
Totally isolated E, V, N conserved	Exchange heat $T \approx \text{const}$ $T, V, N \sim \text{conserved}$ $Z(T, V, N)$ $F = -kT \ln(Z)$ $P = -\frac{\partial F}{\partial V} \quad S = -\frac{\partial F}{\partial T} \quad \mu = \frac{\partial F}{\partial N}$	Exchange heat / particles $T, \mu \approx \text{const}$ $(T, V, \mu) \sim \text{conserved}$ $\mathcal{Z}(T, V, \mu)$ $J = -kT \ln(\mathcal{Z})$ $S = -\frac{\partial J}{\partial T} \quad P = -\frac{\partial J}{\partial V}$ $N = -\frac{\partial J}{\partial \mu}$

In thermo limit the physics at eqbm are $\approx$ the same in all ensembles

$\Rightarrow$ (M) $E = \text{const}$

(G-C) $\langle E \rangle \approx \text{const} = E_F$ most probable value of E

(C) $\langle E \rangle \approx \text{const} = E$

$\langle N \rangle \approx \text{const} \approx N_F$

We will be dealing with quantum systems with discrete energy spectrums

Heisenberg's uncertainty principle $\hbar \neq 0$ (q, p) not appropriate variables since there is no precise values for the trajectory

Microcanonical

$P_n = \frac{1}{\Omega(E, V, N)}$, where n denotes the set of quantum numbers that completely determine a microstate

Example Harmonic Oscillator

A system which contains N one-dimensional h.o with the same frequency

$$\epsilon_i = (n_i + \frac{1}{2}) \hbar \omega$$

$$E = \sum_{i=1}^N \hbar \omega (n_i + \frac{1}{2}) = \frac{1}{2} N \hbar \omega + \hbar \omega \sum_{i=1}^N n_i$$

Ω is effectively the number of all possible combinations of $\{n_i\}$ s.t.

$$\sum_{i=1}^N n_i = (\frac{E}{\hbar \omega} - \frac{N}{2}) \equiv \tilde{E}$$

$$\Rightarrow \Omega(E, V, N) = \sum_{\{n_1, \dots, n_N\} \in N} \delta(\sum_{i=1}^N n_i, \tilde{E})$$

$$W(E, V, N) \Delta E = \int_{\substack{d\mathbf{q} d\mathbf{p} \\ E < H < E + \Delta E}} \frac{d\mathbf{q} d\mathbf{p}}{h^3} \quad (\text{classical})$$

Canonical

In this case, the probability of a given microstate is

$$p_n = \frac{e^{-\beta E_n}}{Z(T, V, N)}, \quad \beta = \frac{1}{kT}$$

$$\sum_n p_n = 1 = \frac{1}{Z} \sum_n e^{-\beta E_n} \Rightarrow Z(T, V, N) = \sum_n e^{-\beta E_n}$$

$$\downarrow \text{(classical)} \quad Z = \int \frac{d\mathbf{q} d\mathbf{p}}{h^3} e^{-\beta H}$$

Here is any function of E , not F should have written $\mathcal{F}$

Recall that we showed:

$$\int_{\text{all phase space}} \frac{d\mathbf{q} d\mathbf{p}}{h^3} e^{-\beta H(\mathbf{q}, \mathbf{p})} \mathcal{F}(\mathbf{q}, \mathbf{p}) = \langle \mathcal{F} \rangle = \int_{\text{all energies}} dE W(E) e^{-\beta E} \mathcal{F}(E)$$

$$\text{intuitively} \Rightarrow \sum_n e^{-\beta E_n} \mathcal{F}(E_n) = \sum_{E=0}^{\infty} \mathcal{F}(E) e^{-\beta E} \Omega(E) = \langle \mathcal{F} \rangle$$

Can we write the above in bra-ket notation?

$$Z = \sum_n e^{-\beta E_n} = \sum_n \langle n | e^{-\beta \hat{H}} | n \rangle \equiv \text{Tr} e^{-\beta \hat{H}} \quad |n\rangle : \text{eigenstates of } \hat{H}$$

$$\hat{\rho} \equiv e^{-\beta \hat{H}} : \text{thermal density operator}$$

$$\langle X \rangle = \sum_n \langle n | e^{-\beta \hat{H}} \hat{X} | n \rangle$$

Grand Canonical

$$P_{\Omega, N} = \frac{1}{\mathcal{Z}} e^{\beta \mu N} \quad \text{replace with discrete sum}$$

$$J = -KT \ln \mathcal{Z} \quad \mathcal{Z} = \sum_{N=0}^{\infty} e^{\beta \mu N} Z(T, V, N)$$

Example Harmonic Oscillators

study the system of Id h.o in the thermodynamic limit.

This implies that we can use our preferred ensemble.
 microcanonical usually more difficult, use C or G first.
 h.o.s are not identical here (but assume ω the same for all)

We will start by considering the partition function.

$$Z(T, V, N) = \sum e^{-\beta E_0}$$

$$\begin{aligned} E_0 &= \sum_{i=1}^N (n_i + \frac{1}{2}) \hbar \omega = \sum_{i=1}^N \epsilon(n_i) \rightarrow Z = \sum_{(n_1, \dots, n_N)} e^{-\beta \sum_{i=1}^N \epsilon(n_i)} \\ &= \sum_{\{n_1, \dots, n_N\}} e^{-\beta \epsilon(n_1)} \dots e^{-\beta \epsilon(n_N)} \\ &= \sum_{n_1} e^{-\beta \epsilon(n_1)} \dots \sum_{n_N} e^{-\beta \epsilon(n_N)} \\ &= \prod_{i=1}^N \left(\sum_{n_i} e^{-\beta \epsilon(n_i)} \right) = \prod_{i=1}^N J_i \end{aligned}$$

$$\text{but } J_i = J = \sum e^{-\beta \epsilon(n)}$$

$$\Rightarrow Z = J^N$$

Now we must compute J for each h.o

$$J = \sum_n e^{-\beta \epsilon(n)} = \sum_n e^{-\beta \hbar \omega (n + \frac{1}{2})} = \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n + \frac{1}{2})} = e^{-\beta \hbar \omega / 2} \sum_{n=0}^{\infty} (e^{-\beta \hbar \omega})^n$$

geometric series ratio $e^{-\beta \hbar \omega}$

$$J = \frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}} \rightarrow Z = \left(\frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}} \right)^N$$

$$= \frac{1}{(e^{\frac{\beta \hbar \omega}{2}} (1 - e^{-\frac{\beta \hbar \omega}{2}}))^N} = \frac{1}{(e^{\frac{\beta \hbar \omega}{2}} - e^{-\frac{\beta \hbar \omega}{2}})^N} = \frac{1}{(2 \sinh(\frac{\beta \hbar \omega}{2}))^N}$$

Note if h.o.s had different ω s $Z = \prod_{i=1}^N \frac{1}{2 \sinh(\frac{\beta \hbar \omega_i}{2})}$

Check against classical limit: $\beta \hbar \omega \ll 1$
 $\hbar \omega \ll kT$
 $\hbar$ smallest } classical limit

$$Z = \frac{1}{2^N \sin^2(\frac{\beta \hbar \omega}{2})} \xrightarrow{\text{classical limit}} \frac{1}{2^N (\frac{\beta \hbar \omega}{2})^N} = (\beta \hbar \omega)^{-N}$$

$$F = -kT \ln Z = \frac{N \hbar \omega}{2} + kT N \ln [1 - e^{-\beta \hbar \omega}]$$

$$\Rightarrow S = kN \left\{ \frac{\beta \hbar \omega}{e^{\beta \hbar \omega} - 1} - \ln(1 - e^{-\beta \hbar \omega}) \right\} \xrightarrow{\text{check classical limit}} kN \{1 - \ln(\beta \hbar \omega)\}$$

$$P = 0 \quad M = \frac{F}{N} \quad C_V = -T \left(\frac{\partial^2 F}{\partial T^2} \right)$$

$$\langle E \rangle = F + TS = N \left\{ \frac{\hbar \omega}{2} + \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} \right\} \xrightarrow{\text{check classical limit}} NkT$$

Note: Quantum oscillators do not satisfy the equipartition of energy theorem only true in classical limit

Now consider the Microcanonical Ensemble

$$\Omega(E, V, N) = \sum_{\{n_1, \dots, n_N\}} \delta(\sum n_i \tilde{E}) \quad \tilde{E} = \left(\frac{E}{\hbar \omega} - \frac{N}{2} \right), \text{ some positive number}$$

We need to find how many ways we can distribute $\tilde{E}$ into N groups $\{n_1, \dots, n_N\}$

Think of $\tilde{E}$ as the number of identical balls & the N groups as the boxes we need to distribute the into (we can have empty boxes "stars & bars" think of where to place the $(N-1)$ walls in $(\tilde{E} + N - 1)$ positions)

$$\rightarrow \binom{\tilde{E} + N - 1}{N - 1} \Rightarrow \Omega = \frac{(\tilde{E} + N - 1)!}{(N - 1)! \tilde{E}!}$$

Another method to compute Ω , if you know Z

$$Z = \sum_n e^{-\beta E(n)} = e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\{n_1, \dots, n_N\}} e^{-\beta \hbar \omega (n_1 + \dots + n_N)}$$

$$= e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\tilde{E}=0}^{\infty} e^{-\beta \hbar \omega \tilde{E}} \sum_{\{n_1, \dots, n_N\}} \delta(\sum n_i \tilde{E}) = e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\tilde{E}=0}^{\infty} e^{-\beta \hbar \omega \tilde{E}} \Omega(\tilde{E}, V, N)$$

$$\text{ux result for } Z \rightarrow \frac{e^{-\frac{\beta \hbar \omega N}{2}}}{(1 - e^{-\beta \hbar \omega})^N} = e^{-\frac{\beta \hbar \omega N}{2}} \sum_{\tilde{E}=0}^{\infty} e^{-\beta \hbar \omega \tilde{E}} \Omega(\tilde{E})$$

$$\sum_{\tilde{E}=0}^{\infty} \frac{1}{(1 - e^{-\beta \hbar \omega})^N} = \sum_{\tilde{E}=0}^{\infty} x^{\tilde{E}} \Omega(\tilde{E}) \xrightarrow{\text{Exercise}} \left(\frac{\tilde{E} + N - 1}{N - 1} \right)$$

StatPhys 2 Sound Wave Propagation in a Solid 30/01

- Two ways to describe
- 1) Consider as a collection of H.O.s with given frequencies
 - 2) Consider as a gas, collection of free particles $\equiv$ "phonons"

$$1) H = V_0 + \sum_{i=1}^{3N} \frac{1}{2} (m \omega_i^2 q_i^2 + \frac{1}{m} p_i^2)$$

(rest energy)

$$QM \rightarrow E_{tot} = V_0 + \sum_{i=1}^{3N} (n_i + \frac{1}{2}) \hbar \omega_i$$

interested in the thermo-limit
so we use canonical ensemble

$$Z = e^{-\beta V_0} \prod_{i=1}^{3N} Z_i, \quad Z_i = \frac{e^{-\frac{1}{2} \beta \hbar \omega_i}}{1 - e^{-\beta \hbar \omega_i}}$$

$$\rightarrow Z = e^{-\beta V_0} \prod_{i=1}^{3N} \frac{e^{-\frac{1}{2} \beta \hbar \omega_i}}{1 - e^{-\beta \hbar \omega_i}}$$

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z$$

$$= -\frac{\partial}{\partial \beta} (-\beta V_0 + \sum_{i=1}^{3N} \ln Z_i)$$

$$= -\frac{\partial}{\partial \beta} (-\beta V_0 - \frac{\beta}{2} \sum_{i=1}^{3N} \hbar \omega_i - \sum_{i=1}^{3N} \ln (1 - e^{-\beta \hbar \omega_i}))$$

$$= V_0 - \frac{\hbar}{2} \sum_{i=1}^{3N} \omega_i - \sum_{i=1}^{3N} \frac{\hbar \omega_i e^{-\beta \hbar \omega_i}}{1 - e^{-\beta \hbar \omega_i}}$$

independent of β
 $\Rightarrow$ "ground state"

$$\beta \hbar \omega_i = \frac{\hbar \omega_i}{kT} \text{ classical}$$

$\gg 1 \Rightarrow$ Quantum

$$\langle E \rangle \approx V_0 + \sum_{i=1}^{3N} \frac{\hbar \omega_i}{2}$$

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,N} = K \sum (\beta \hbar \omega_i)^2 \frac{e^{-\beta \hbar \omega_i}}{(e^{-\beta \hbar \omega_i} - 1)^2}$$

Assume $\omega_i \approx \omega_E$

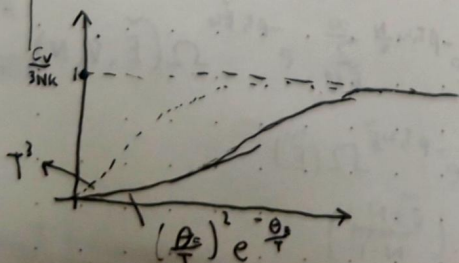
$$\rightarrow \langle E \rangle = 3N \hbar \omega_E \left(\frac{1}{2} + \frac{e^{-\beta \hbar \omega_E}}{1 - e^{-\beta \hbar \omega_E}} \right) + V_0$$

$$C_V = 3Nk \frac{x^2 e^x}{(e^x - 1)^2} = 3Nk \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\frac{\theta_E}{T}}}{(e^{\frac{\theta_E}{T}} - 1)^2}$$

$x = \beta \hbar \omega_E$ $x = \frac{\theta_E}{T}$

$$\text{let } x \gg 1 \quad C_V \approx 3Nk \left(\frac{\theta_E}{T} \right)^2 e^{-\theta_E/T}$$

$kT \ll \hbar \omega_E$
 $T \ll \theta_E$



$$C_V \sim T^3, \quad \langle E \rangle \sim T^4, \quad \ln Z \sim -\frac{1}{\beta^3}$$

$$\text{we have } \ln Z = -\left(\beta \hbar \sum_{i=1}^{3N} \omega_i + \beta P V_0\right) - \sum_{i=1}^{3N} \ln(1 - e^{-\beta \hbar \omega_i})$$

Vibrational Modes in a Solid

$$C_V \sim T^3 \text{ for all materials}$$

$$\ln Z = -\beta \hbar \sum_{i=1}^{3N} (\omega_i + V_0) - \sum_{i=1}^{3N} \ln(1 - e^{-\beta \hbar \omega_i})$$

ω_0 scale,
 $\omega_i > \omega_0$ don't contribute to C_V

$$g(\omega) = \sum_{i=1}^{3N} \int_0^\infty \delta(\omega - \omega_i) d\omega = \sum_{i=1}^{3N} 1 = 3N$$

$$\text{Approximate } \tilde{g}(\omega) : \int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N \Rightarrow g(\omega) \approx 0 \text{ for } \omega > \omega_0, \text{ assume continuous}$$

a good estimate for $\omega_0 \sim \frac{1}{a}$, $a \equiv$ interatomic distance

Waves with $\lambda \gg a$ would not propagate through the material
can think of $g(\omega)$ as an almost continuous spectrum of frequencies

$$\ln Z = -\beta \hbar \sum_{i=1}^{3N} (\omega_i + V_0) - \int_0^{\omega_0} d\omega g(\omega) \ln(1 - e^{-\beta \hbar \omega})$$

Estimate $\tilde{g}(\omega)$, Investigate homo properties of $\tilde{g}(\omega)$

$$\text{Volume fixed } \simeq N a^3 \stackrel{a \rightarrow \frac{1}{\omega_0}}{=} \frac{N L^3}{N} \left(\frac{a}{L}\right)^3, \quad \omega_0 \sim \frac{1}{a} \rightarrow \omega'_0 \simeq \omega_0$$

$$1) \int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N$$

$$2) \text{ Rescale: } \int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N L^3 \Leftrightarrow \int_0^{\omega'_0} d\tilde{\omega} \tilde{g}(\tilde{\omega}) = 3N L^3$$

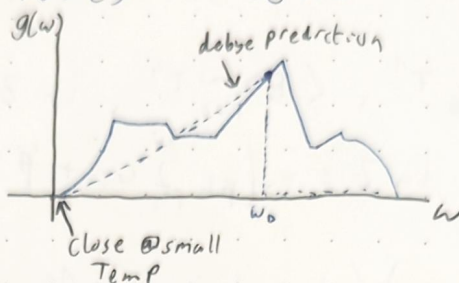
$$\int_0^{\omega_0} \tilde{g}(\omega) d\omega = 3N L^2 \Leftrightarrow \int_0^{\omega'_0} d\omega \tilde{g}(\omega) = 3N L^2$$

$$\Leftrightarrow \int_0^{\omega'_0} \tilde{g}(\omega) d\omega = \int_0^{\omega'_0} L^2 \tilde{g}(\omega) d\omega$$

$$\Rightarrow \tilde{g}(L\omega) = L^2 \tilde{g}(\omega) \Rightarrow \omega \text{ homo degree } 2$$

$$\Rightarrow \tilde{g}(\omega) = A \omega^2 \quad (\text{Euler})$$

$$\tilde{g}(\omega) = \begin{cases} 0, & \omega > \omega_D \\ \frac{a}{\omega_D} \left(\frac{\omega}{\omega_D}\right)^2, & \omega \leq \omega_D \end{cases}$$



$$\langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z$$

$$= V_0 + \frac{\hbar}{2} \int_0^\infty d\omega g(\omega) \omega + \int_0^\infty d\omega \frac{g(\omega) \hbar \omega}{e^{\beta \hbar \omega} - 1}$$

$$\ln Z = -\beta V_0 - \frac{\beta \hbar}{2} \int_0^\infty d\omega g(\omega) \omega - \int_0^\infty d\omega g(\omega) \ln(1 - e^{-\beta \hbar \omega})$$

$$C_V = \frac{\partial \langle E \rangle}{\partial T} = k \int_0^\infty d\omega g(\omega) \frac{(\beta \hbar \omega)^2}{(e^{\beta \hbar \omega} - 1)^2} e^{\beta \hbar \omega} = k \int_0^{\omega_D} d\omega \left(\frac{9N}{\omega_D^3}\right) \omega^2 \frac{(\beta \hbar)^2 \omega^2}{(e^{\beta \hbar \omega} - 1)^2} e^{\beta \hbar \omega}$$

$$= k \left(\frac{9N}{\omega_D^3}\right) (\beta \hbar)^2 \int_0^{\omega_D} d\omega \frac{\omega^4 e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2} = \left(\frac{9N}{\omega_D}\right) \frac{\hbar^2}{k T^2} \int_0^{\omega_D} d\omega \frac{\omega^2 e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2}$$

$$= \frac{\hbar^2}{k T^2} \frac{9N}{\omega_D^3} \left(\frac{k T}{\hbar}\right)^5 \int_0^{\theta_D} dx \frac{x^4 e^x}{(e^x - 1)^2} \quad \theta_D = \frac{\hbar \omega_D}{T}$$

$$C_V = 9Nk \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D} dx \frac{x^4 e^x}{(e^x - 1)^2}$$

for $\beta \hbar \omega \gg 1$ (small T):

$$C_V = 9Nk \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D} dx \frac{x^4 e^x}{(e^x - 1)^2} \quad \leftarrow \text{some number as long as it converges}$$

$\Rightarrow C_V \sim T^3$ as expected

for large $T \gg \theta_D$:

$$C_V = 9Nk \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D} dx \frac{x^4}{(1-x)^2} = 9Nk \left(\frac{T}{\theta_D}\right)^3 \left(\frac{\theta_D}{T}\right)^3 \cdot \frac{1}{3}$$

$= 3Nk \rightarrow$ classical result

Stat Phys 2

Vibrational Modes in Solid 01/02

Computation of $I = \int_0^\infty dx \frac{x^4 e^x}{(e^x - 1)^2} = - \frac{x^4}{e^x - 1} \Big|_{x=0}^\infty - 4 \int_0^\infty dx \frac{x^3}{e^x - 1}$

$\Rightarrow \int_0^\infty dx \frac{x^3}{e^x - 1} = \sum_{n=0}^\infty \int_0^\infty dx x^3 e^{-(n+1)x} \stackrel{\text{I.B.P.}}{=} \sum_{n=0}^\infty \frac{6}{(n+1)^4} = \frac{\pi^4}{15}$

used: $\frac{1}{e^x - 1} = \frac{e^{-x}}{1 - e^{-x}} = e^{-x} \sum_{n=0}^\infty e^{-nx}$

Stat Phys 2 Theory of Ideal Systems of Identical Particles

13/02

$$E = \epsilon_{i_1} + \epsilon_{i_2} + \dots + \epsilon_{i_N} \text{ No longer makes sense}$$

$$E = \sum_i n_i \epsilon_i \quad \epsilon_i: \text{distinct energy level}$$

$$N = \sum_i n_i$$

"occupation number":
how many particles
occupy each level

Suppose we want to work in canonical ensemble

$$T \text{ or } \beta, V, N$$

$$Z = \sum_{\{n_i, \dots, n_{\infty}\}} e^{-\beta \sum_i n_i \epsilon_i} \delta_{N, \sum_i n_i} = \begin{cases} 1, & N = \sum_i n_i \\ 0, & \text{otherwise} \end{cases}$$

So use Grand-canonical ensemble

$$\mathcal{Z}(T, V, \mu) = \sum_{N=0}^{\infty} e^{\beta \mu N} Z(T, V, N) = \sum_{N=0}^{\infty} e^{\beta \mu N} \sum_{\{n_i, \dots\}} e^{-\beta \sum_i n_i \epsilon_i} \delta_{N, \sum_i n_i} =$$

$$= \sum_{\{n_i, \dots\}} e^{-\beta \sum_i n_i \epsilon_i} \sum_{N=0}^{\infty} e^{\beta \mu N} \delta_{N, \sum_i n_i} = \sum_{\{n_i, \dots\}} e^{-\beta \sum_i n_i (\epsilon_i - \mu)}$$

$$\mathcal{Z} = \sum e^{-\beta \sum_i n_i (\epsilon_i - \mu)} = \sum_{n_1} \sum_{n_2} \dots e^{-\beta n_1 (\epsilon_1 - \mu)} e^{-\beta n_2 (\epsilon_2 - \mu)} \dots$$

$$\Rightarrow \mathcal{Z} = \prod_i \left(\sum_{n_i} e^{-\beta n_i (\epsilon_i - \mu)} \right) \stackrel{\text{ideal}}{\Rightarrow} \mathcal{Z} = \prod_i \mathcal{Z}_i$$

$\mathcal{Z}_i$: grand-canonical ensemble of one level i

Two Cases: ① A single energy level can be populated by any number of particles
 $n_i = 0, 1, \dots$
bosons, particles of zero or integer spin

② A single energy level can be populated by at most one particle
Fermi: Dirac statistics
Fermions, half integer spin

13/02

Ideal systems of identical particles Stat Phys

$$\delta_i = \begin{cases} \textcircled{1} \sum_{n_i=0}^{\infty} e^{-\beta n_i (\epsilon_i - \mu)} = \frac{1}{1 - e^{-\beta (\epsilon_i - \mu)}}, & \epsilon_i \geq \mu \\ \textcircled{II} \sum_{n_i=0}^1 e^{-\beta n_i (\epsilon_i - \mu)} = 1 + e^{-\beta (\epsilon_i - \mu)}, & \epsilon_i < \mu \end{cases}$$

$$\mathcal{Z}(T, V, \mu) = \begin{cases} \textcircled{1} \prod_i \frac{1}{1 - e^{-\beta (\epsilon_i - \mu)}}, & \epsilon_i \geq \mu \\ \textcircled{II} \prod_i (1 + e^{-\beta (\epsilon_i - \mu)}) \end{cases}$$

$$J \equiv -kT \ln \mathcal{Z} = \begin{cases} \textcircled{1} kT \sum_i \ln(1 - e^{-\beta (\epsilon_i - \mu)}) \\ \textcircled{II} -kT \sum_i \ln(1 + e^{-\beta (\epsilon_i - \mu)}) \end{cases} \rightarrow J = \pm kT \sum_i \ln(1 \pm e^{-\beta (\epsilon_i - \mu)})$$

upper signs $\rightarrow$ bosons

(J(T, V, μ))

fugacity $\rightarrow$

$$J(T, V, z \equiv e^{\beta \mu}) = \pm kT \sum_i \ln(1 \pm z e^{-\beta \epsilon_i / kT})$$

characteristics

$$\text{average occupation number: } \langle n_i \rangle = \sum_{\tilde{n}_i} \overbrace{\text{pr}(\epsilon_i, \tilde{n}_i)}^{\text{characteristics}} \cdot n_i$$

$$\text{we know } \text{pr}(\tilde{E}, N) = \frac{e^{-\beta \sum_i n_i (\epsilon_i - \mu)}}{\mathcal{Z}}$$

$$\text{pr}(\epsilon_i, \tilde{n}_i) = \frac{1}{\mathcal{Z}} \sum_{\{n_1, \dots, n_i, \dots, n_i = \tilde{n}_i\}} e^{-\beta n_1 (\epsilon_1 - \mu)} e^{-\beta n_i (\epsilon_i - \mu)}$$

$$= \frac{1}{\mathcal{Z}} \prod_{\substack{\text{all } \epsilon\text{-levels} \\ \text{except } i\text{th}}} \left(\sum_{n_j} e^{-\beta n_j (\epsilon_j - \mu)} \right) e^{-\beta \tilde{n}_i (\epsilon_i - \mu)}$$

$$= \frac{e^{-\beta \tilde{n}_i (\epsilon_i - \mu)}}{\delta_i}$$

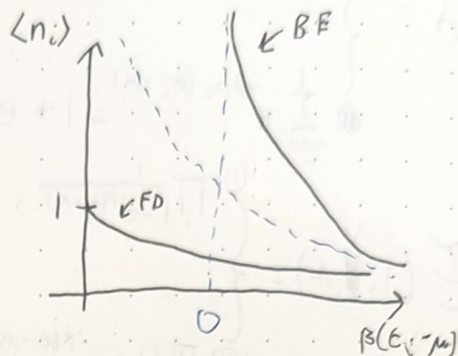
$$\Rightarrow \langle n \rangle = \sum_{\tilde{n}_i} \frac{e^{-\beta n_i (\epsilon_i - \mu)}}{\delta_i} n_i \stackrel{\text{exercise}}{=} \sum_{\tilde{n}_i} \frac{\partial}{\partial \beta} \ln \delta_i$$

fugacity

Stat Phys 2 Ideal Systems of Identical Particles 14/02

$$\langle n_i \rangle = z \frac{\partial}{\partial z} \ln \bar{\mathcal{Z}}_i$$

$$z \equiv e^{\beta \mu}$$



$$J = \pm \frac{1}{\beta} \sum_i \ln(1 \mp z e^{-\beta \epsilon_i})$$

$$\bar{\mathcal{Z}} = \prod_i (1 \mp z e^{-\beta \epsilon_i})^{\pm 1}$$

$$\bar{\mathcal{Z}} = (1 \mp z e^{-\beta \epsilon_i})^{(\mp 1)}$$

$$\ln \bar{\mathcal{Z}}_i = \mp \ln(1 \mp z e^{-\beta \epsilon_i})$$

$$\rightarrow \langle n_i \rangle = \frac{e^{-\beta(\epsilon_i - \mu)}}{1 \mp e^{-\beta(\epsilon_i - \mu)}} \Leftrightarrow \langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i - \mu)} \mp 1}$$

Classical regime

$$\beta(\epsilon_i - \mu) \gg 1 \rightarrow \langle n_i \rangle_{FD} \sim e^{-\beta(\epsilon_i - \mu)} \quad \left. \begin{array}{l} \text{Maxwell-} \\ \text{Boltzmann} \end{array} \right\}$$

$$\langle n_i \rangle_{BE} \sim e^{-\beta(\epsilon_i - \mu)}$$

Classical $\Rightarrow \beta \epsilon_i \ll 1$ This shows that $\beta \mu \ll -1$ to compete

$$\text{or equivalently } z \equiv e^{\beta \mu} \ll 1$$

So $z \ll 1$ is used to identify classical regime in these systems

(BE): We assumed $\beta(\epsilon_i - \mu) > 0$, it is justified by disallowing $\langle n_i \rangle < 0$
 $\langle n_i \rangle \rightarrow \infty$ for $\beta(\epsilon_i - \mu) \rightarrow 0$

The condition $\mu < \epsilon_i$, really $\mu < \epsilon_0 \equiv \text{ground state}$

(FD): $0 < \langle n_i \rangle < 1$

positive $\beta(\epsilon_i - \mu) \rightarrow \langle n_i \rangle \rightarrow 0$

negative $\beta(\epsilon_i - \mu) \rightarrow \langle n_i \rangle \rightarrow 1$

$\beta(\epsilon_i - \mu) \gg 1$

$$S = - \left(\frac{\partial J}{\partial T} \right)_{\mu, V}, \quad P = - \left(\frac{\partial J}{\partial V} \right)_{T, \mu}, \quad N = - \left(\frac{\partial J}{\partial \mu} \right)_{T, V}$$

$$\langle E \rangle = \frac{\partial \ln \bar{\mathcal{Z}}}{\partial \beta} \bigg|_{\mu, V}, \quad \langle N \rangle = z \frac{\partial \ln \bar{\mathcal{Z}}}{\partial z} \bigg|_{\beta, V}, \quad \langle E \rangle = \frac{\partial \ln \bar{\mathcal{Z}}}{\partial \beta} \bigg|_{\mu, V}$$

0: Exprove these are equivalent

Stat Phys Ideal, non relativistic bosonic gas

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In volume V

assume bosonic

$$\epsilon_i = \frac{\hbar^2 i^2}{2mL^2}, i=0, \pm 1, \pm 2, \dots$$

$$\mathcal{Z} = \prod_i (1 - e^{-\beta(\epsilon_i - \mu)})^{-1} \quad \mu < \epsilon_0 = 0$$

$L \sim$ characteristic scale of the vessel

$$V \sim L^3$$

$$\langle n_i \rangle = \frac{1}{e^{\beta(\epsilon_i - \mu)} - 1} \Rightarrow \langle n_0 \rangle = \frac{z}{1-z} \xrightarrow{\mu \rightarrow 0} \infty$$

$$0 < z \leq e^{\beta\mu} < 1$$

$$\ln \mathcal{Z} = - \sum_i \ln(1 - e^{-\beta(\epsilon_i - \mu)}) = - \int d\epsilon W(\epsilon) \ln(1 - z e^{-\beta\epsilon})$$

$$W(\epsilon) = \sum_i \delta(\epsilon - \epsilon_i)$$

single particle density of states

We will treat the problem approximately semi-classically

$$W(\epsilon) = \delta(\epsilon) + \sum_{i \neq 0} \delta(\epsilon - \epsilon_i) \underset{\substack{\text{dense} \\ \text{spectrum}}}{\approx} \delta(\epsilon) + (\text{deg.}) \int_{-\infty}^{\infty} d^3i \delta(\epsilon - \frac{\hbar^2 i^2}{2mL^2})$$

$$\left. \begin{aligned} p = \hbar k, \quad k_x &\approx \frac{2\pi i_x}{L} \\ k_y &\approx \frac{2\pi i_y}{L} \\ k_z &\approx \frac{2\pi i_z}{L} \end{aligned} \right\} \rightarrow \text{similar for } y, z \quad p_x = \frac{\hbar i_x}{L}$$

with this identification

$$\begin{aligned} W(\epsilon) &\approx \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} \int_{-\infty}^{\infty} d^3p \delta(\epsilon - \frac{p^2}{2m}) \\ &= \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} (4\pi) \int_0^{\infty} dp p^2 \delta(\epsilon - \frac{p^2}{2m}) \\ &\stackrel{t = \frac{p^2}{2m}}{\approx} \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} (2\pi) (2m)^{3/2} \int_0^{\infty} dt t^{1/2} \delta(\epsilon - t) \\ &= \delta(\epsilon) + (\text{deg.}) \frac{V}{h^3} (2\pi) \frac{(2m\epsilon)^{3/2}}{\epsilon} \end{aligned}$$

$$W_c(\epsilon, V, 1)$$

↑ classical

$$\ln \mathcal{Z} = - \int_{\text{all energies}} d\epsilon W(\epsilon) \ln(1 - z e^{-\beta\epsilon})$$

$$\approx - \int_0^{\infty} d\epsilon (\delta(\epsilon) + W_c(\epsilon)) \ln(1 - z e^{-\beta\epsilon})$$

$$= - \int_{-\epsilon}^{\infty} d\epsilon \delta(\epsilon) \ln(1 - z e^{-\beta\epsilon}) - \int_0^{\infty} d\epsilon W_c(\epsilon) \ln(1 - z e^{-\beta\epsilon})$$

$$= -\ln(1-z) - \int_0^{\infty} d\epsilon W_c(\epsilon) \ln(1 - z e^{-\beta\epsilon}) \Rightarrow$$

$$\ln \mathcal{Z} = -\ln(1-z) - \frac{\text{deg.} V (2\pi) (2m)^{3/2}}{h^3} \int_0^{\infty} d\epsilon \epsilon^{1/2} \ln(1 - z e^{-\beta\epsilon})$$

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Ideal, Non rel boson gas Stat Phys 2

$$N \equiv \text{fixed} = \langle N \rangle = \langle n_0 \rangle + \langle N_{\text{ex}} \rangle$$

for canonical ensemble, this gives $M = M(\beta, V, N)$

$$\langle N_{\text{ex}} \rangle \propto \text{Li}_{5/2}(z), \quad 0 < z < 1 \quad \text{polylogs monotonically increase}$$

Study the system in Thermo limit $\{N \gg 1\}$ with the Canonical Ensemble

$$\langle N \rangle \stackrel{\text{GCE}}{=} f(p, V, m) \stackrel{\text{C.E.}}{=} M = g(p, V, N \equiv \langle N \rangle)$$

$$\begin{array}{c} \text{fixed} \\ \downarrow \\ N = \langle n_0 \rangle + \langle N_{\text{ex}} \rangle \end{array} \begin{array}{c} \text{exchange} \\ \downarrow \end{array} \rightarrow \langle N_{\text{ex}} \rangle = N - \langle n_0 \rangle = (\text{deg}) V \left(\frac{2\pi m}{h^2 \beta} \right)^{3/2} \text{Li}_{5/2}(z)$$

$\text{Li}_{5/2}(z)$ mon increasing as z a number

$$\Rightarrow 0 < z < 1, \quad \text{Li}_{5/2}(z) < \text{Li}_{5/2}(1)$$

$$\rightarrow \langle N_{\text{ex}} \rangle \leq (\text{deg}) V \left(\frac{2\pi m}{h^2 \beta} \right)^{3/2} \text{Li}_{5/2}(1)$$

$$\text{Canonical Ensemble, for } N > N_{\text{ex}} \Rightarrow T < \underbrace{\left(\frac{h^2}{2\pi m} \left(\frac{N}{V} \right)^{2/3} \left(\frac{1}{(\text{deg}) \text{Li}_{5/2}(1)} \right)^{2/3} \right)}_{T_{\text{critical temp}}}$$

$$\text{When } T \ll T_c, \quad z = \frac{\langle n_0 \rangle}{1 - \langle n_0 \rangle} = \frac{1}{\frac{1}{\langle n_0 \rangle} - 1} \approx 1 - \frac{1}{\langle n_0 \rangle}$$

$$z \rightarrow 1 \Leftrightarrow \langle n_0 \rangle \gg 1$$

$$T/T_c \ll 1$$

$$\langle N_{\text{ex}} \rangle = N - \langle n_0 \rangle = N \left(\frac{T}{T_c} \right)^{3/2} \frac{\text{Li}_{5/2}(z)}{\text{Li}_{5/2}(1)} \stackrel{z \rightarrow 1}{\approx} N \left(\frac{T}{T_c} \right)^{3/2}$$

$$\langle n_0 \rangle \approx N \left(1 - \left(\frac{T}{T_c} \right)^{3/2} \right), \quad \frac{T}{T_c} \ll 1 \text{ or } z \rightarrow 1$$

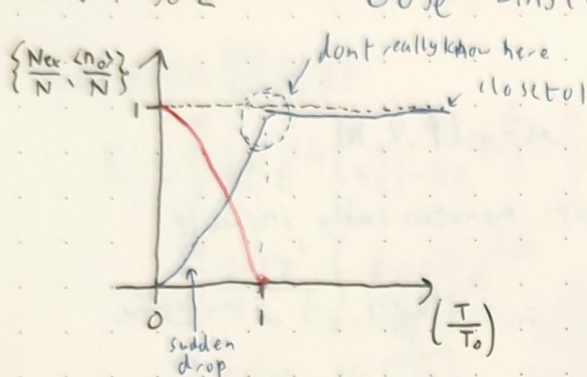
B-E condensation, low temperatures } Quantum Physics
 $z \rightarrow 1$
 bound on N_{ex}

$$T \gg T_c \Rightarrow \text{expect classical results: } N_{\text{ex}} = N \left(\frac{T}{T_c} \right)^{3/2} \frac{\text{Li}_{5/2}(z)}{\text{Li}_{5/2}(1)} \approx N \left(\frac{T}{T_c} \right)^{3/2} \frac{z}{\text{Li}_{5/2}(1)}$$

$$\text{Li}_{5/2}(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^{5/2}} \approx z$$

Note that this result is constant

$$\text{with } \langle n_0 \rangle \ll 1, \quad N \approx N_{\text{ex}} \Rightarrow \left(\frac{T}{T_c} \right)^{3/2} \approx \frac{1}{z} \gg 1$$



At $T = T_c$ we see a new phase.

BE condensation is a phase transition

Equation of State

$$\frac{z}{1-z} = \langle n_0 \rangle$$

$$PV = NkT \text{ classically, } = \frac{2}{3} \langle E \rangle$$

$$J = -kT \ln \Xi = -P \cdot V, \quad \langle E \rangle = - \left. \frac{\partial \ln \Xi}{\partial \beta} \right|_{V, z} \left. \vphantom{\frac{\partial \ln \Xi}{\partial \beta}} \right\} \Rightarrow P \cdot V = \frac{2}{3} \langle E \rangle - kT \ln(1-z)$$

$$= \frac{2}{3} \langle E \rangle + kT \ln(1 + \langle n_0 \rangle)$$

Exercise

probability for

to have

What is the average no of particles with energies between E & $E + dE$

$$\langle n(E) \rangle$$

exact average occupation number

$$\langle N_{ex} \rangle = a V \beta^{-3/2} \text{Li}_{3/2}(z)$$

$$d\langle N \rangle = \langle n(E) \rangle W(1, E) dE$$

$$\int_0^\infty d\langle N \rangle \approx \langle N_{ex} \rangle = a V \beta^{-3/2} \text{Li}_{3/2}(z)$$

$$P(E) dE = \frac{\langle n(E) \rangle}{\langle N_{ex} \rangle} W_c(1, E) dE \quad (\text{check: } \int_0^\infty P(E) dE = 1)$$

$$\Rightarrow P(E) dE = \frac{\langle n(E) \rangle}{\text{Li}_{3/2}(z)} \frac{z}{\sqrt{\pi}} E^{1/2} \beta^{3/2} dE$$

Exercise: What is the average energy level of the system?

$$\langle E \rangle = \int_0^\infty dE E P(E) = \frac{z}{\sqrt{\pi}} \beta^{3/2} \frac{1}{\text{Li}_{3/2}(z)} \underbrace{\int_0^\infty dE \langle n(E) \rangle E^{3/2}}_I$$

$$\text{Compute I: } \int_0^\infty dE \frac{E^{3/2}}{e^{\beta E} - 1} = \int_0^\infty \frac{dE E^{3/2}}{\frac{e^{\beta E}}{E} - 1} \stackrel{t = \beta E}{=} \beta^{-5/2} \int_0^\infty dt \frac{t^{3/2}}{e^t - 1} = \beta^{-5/2} \frac{\Gamma(5/2)}{\Gamma(5/2)} \text{Li}_{5/2}(z)$$

Stat Phys 2 Ideal N-R Bosonic Gas

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$$\rightarrow \langle E \rangle = \left(\frac{2}{\sqrt{\pi}} \Gamma\left(\frac{5}{2}\right) \right) \frac{Li_{5/2}(z)}{Li_{3/2}(z)} \beta^{-1}$$

~~Exercise~~ Compute $C_{v,N}$ for $T \approx T_c$ ($T \rightarrow T_c$) for a N-R Bosonic Gas

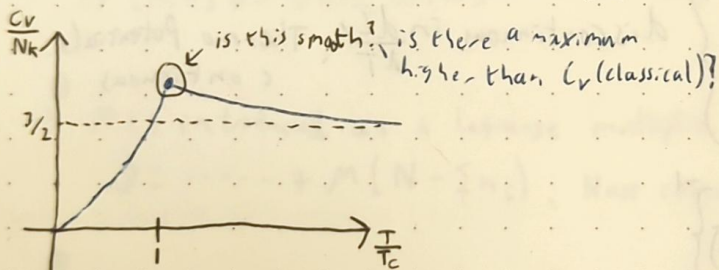
$$C_v = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_{v,N} \quad \text{For } T \approx T_c, z \approx N, N \approx N_{ex} \sim a V \rho_c^{-3/2} Li_{3/2}(1)$$

$$\xrightarrow{\text{sub into } \langle E \rangle} C_v = \frac{15}{4} \frac{Li_{5/2}(1)}{Li_{3/2}(1)} \left(\frac{T}{T_c} \right)^{3/2} \propto 1.925 \left(\frac{T}{T_c} \right)^{3/2}$$

* Compute C_v for $T \gg T_c$ ($z \ll 1$) to show $C_v = \frac{3}{2} N k_B$

$$z \ll 1, Li_{5/2}(z) \approx z \quad T \gg T_c, N \approx N_{ex} \approx a V \rho_c^{-3/2} z$$

$$\xrightarrow{\text{sub into } \langle E \rangle} \langle E \rangle \approx \frac{3}{2} N \beta^{-1} \Rightarrow C_v \approx \frac{3}{2} k_B N$$



What happens to C_v as $T \rightarrow T_c^+$

$$C_{v,N} = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_{v,N} = \frac{-15}{4} a \rho_c^{-3/2} Li_{5/2}(z) + \frac{3}{2} a \rho_c^{-5/2} Li_{3/2}(z) \frac{d \ln z}{dT} = \left. \frac{\partial \langle E \rangle}{\partial T} \right|_{v,N}$$

$$\text{start } N \approx N_{ex} \approx a k^{3/2} T^{3/2} Li_{3/2}(z) \text{ for } T > T_c$$

$$z \frac{d Li_{3/2}(z)}{dz} = Li_{1/2}(z)$$

$$\text{canonical ensemble, } N \text{ fixed} \rightarrow \frac{dN}{dT} = 0 \Leftrightarrow a k^{3/2} \frac{3}{2} T^{1/2} Li_{3/2}(z) + a k^{3/2} T^{3/2} \frac{Li_{3/2}(0)}{z} \left(\frac{dz}{dT} \right) = 0$$

$$\Rightarrow \left(\frac{dz}{dT} \right)_{v,N} \approx - \frac{3}{2T} \left(\frac{z Li_{3/2}(z)}{Li_{1/2}(z)} \right) \text{ how does this behave for } z \rightarrow 1 \quad \int_0^\infty dx \frac{x^{-1/2}}{(e^x - 1)}$$

For $z \rightarrow 1$, the integral in the polylog diverges close to $x \approx 0$, so low values of x give the dominant contribution so $z \sim 1$

$$\{ \quad L_{1/2}(z) \stackrel{z \rightarrow 1}{\sim} \frac{1}{\Gamma(1/2)} \int_0^{\infty} \frac{dx x^{-1/2}}{(x+1)z - 1} + \text{"finite terms"} = \frac{z}{\Gamma(1/2)} \int_0^1 \frac{x^{-1/2}}{(1-z+x)} + \text{"finite terms"} \\ \stackrel{u \equiv \sqrt{x}}{=} \frac{z z(1-z)^{-1/2}}{\Gamma(1/2)} \int_0^1 \frac{du}{1+u^2} + \text{"finite terms"}$$

$$\rightarrow L_{1/2}(z) \propto_{z \rightarrow 1} z(1-z)^{-1/2} \quad \text{and} \quad \frac{dz}{dT} \underset{z \rightarrow 1}{\approx} -\frac{z}{2T} \sqrt{1-z} \frac{N}{ak^{3/2} T^{3/2}} \stackrel{\text{known}}{\approx} A \frac{\sqrt{1-z(T)}}{T^{5/2}}$$

$\rightarrow 1^{\text{st}}$ order ODE for $z = z(T)$, $T \rightarrow T_c^+$

$$z(T_c) = 1 \quad \text{leads to} \quad z(T) \underset{\text{known}}{\approx} 1 - b \left(\frac{T - T_c}{T_c} \right)^2$$

$$\xrightarrow{\text{substit } C_v} \frac{C_v}{kN} \underset{T \rightarrow T_c^+}{\approx} \frac{15}{4} \frac{L_{3/2}(z)}{L_{3/2}(1)} \left(\frac{T}{T_c} \right)^{3/2} - \frac{9}{4} \cdot \# \cdot \left(\frac{T - T_c}{T_c} \right)^2$$

So: ① C_v smoothly approaches T_c as $T \rightarrow T_c^+$

$$\left. \begin{aligned} \text{② } \frac{dC_v}{dT} \Big|_{T \rightarrow T_c^+} &= \left(\dots \right)_{1^{\text{st}} \text{ terms}} + \# \\ \frac{dC_v}{dT} \Big|_{T \rightarrow T_c^-} &= \left(\dots \right)_{1^{\text{st}} \text{ terms}} \end{aligned} \right\} \begin{array}{l} \text{discontinuous in } \frac{dC_v}{dT}, \text{ Thermo Potentials} \\ \text{continuous} \end{array}$$

$$z = \left(1 + \frac{1}{L_{3/2}} \right)_{T \rightarrow T_c}^{-1} \left\{ 1 + \frac{1}{N \left(1 - \left(\frac{T}{T_c} \right)^{3/2} \right)} \right\}^{-1}$$

$$\frac{dz}{dT} \Big|_{N, N} \underset{T \rightarrow T_c}{\sim} \cancel{O(1/N)} O\left(\frac{1}{N}\right) \quad \frac{dz}{dT} \Big|_{N, N} \underset{T \rightarrow T_c}{\sim} O(1)$$

The discontinuity arises because we consider $N \gg 1$ & study the system in the leading $N \rightarrow \infty$ approx. (Thermo Limit)

Comment about what is and is not B-E condensate

Consider the gas, $\epsilon_1 - \epsilon_0 \approx \frac{\hbar^2}{2mV^{2/3}}$, so $kT < \epsilon_1 - \epsilon_0$, then particles will both $i=0$ state but this is not BEC.

Macroscopic $\rightarrow \frac{\hbar^2}{2mV^{2/3}} \ll kT \ll kT_c$ & particles will still populate the groundstate much more

Stat Phys 2 Photon Gas

21/02

Semi-classical treatment of "blackbody" radiation aka "photon gas"

A system of bosonic particles, which are free and can be treated with the methods discussed so far

* Relativistic Particles ($m=0$)

** Number of Photons is not conserved $\Rightarrow \mu=0$

* $\Rightarrow N$ not conserved $\Rightarrow$ no work associated with change of N $\mu=0$

$$\hat{\rho} = \frac{e^{-\beta H}}{\mathcal{Z}} \quad \text{density matrix}$$

Canonical ensemble

generality

$$\mathcal{Z} = e^{\beta \mu N} = 1 \text{ bec}$$

$$\hat{\rho} = \frac{e^{-\beta(\hat{H} - \mu \hat{N})}}{\mathcal{Z}} \quad \text{grand canonical ensemble}$$

$$[\hat{\rho}, \hat{H}] = 0$$

$$\text{conserved} \Leftrightarrow [\hat{N}, \hat{H}] = 0$$

$$\text{if } [\hat{N}, \hat{H}] \neq 0 \text{ then } [\hat{\rho}, \hat{H}] \neq 0 \text{ unless } \mu=0$$

* $\Rightarrow \mu$ is introduced as a Lagrange multiplier in the relevant thermodynamic potential

$$\Phi = \dots + \mu(N - \sum n_i) \quad N \text{ not conserved so no reason to define } \mu$$

* $\Rightarrow$ If N is not conserved, then F or G should not depend on $N \Rightarrow \mu \propto \left(\frac{\partial F}{\partial N}\right)_{TV} = 0$

relativistic means
density of states is what matters

$$W_c(1, \epsilon) \quad \ln(\mathcal{Z}) = \int d\epsilon W_c(1, \epsilon) \underbrace{\ln(1 - \mathcal{Z} e^{-\beta \epsilon})}_{\langle n \rangle}$$

$$W_c(1, p) dp = (\text{deg}) 4\pi \frac{V}{h^3} p^2 dp = (\text{deg}) \frac{4\pi V}{h^3} (\epsilon^2 - m^2) \cdot \frac{2\epsilon d\epsilon}{2\sqrt{\epsilon^2 - m^2}} = (\text{deg}) \frac{4\pi V}{h^3} \sqrt{\epsilon^2 - m^2} \epsilon d\epsilon$$

set $c=1 \rightarrow \epsilon = \sqrt{p^2 + m^2}$

For photons: $(\text{deg}) = 2$ (2 polarisations) Take as given for this module
 $m=0$

$$\text{then } W_c(1, \epsilon) d\epsilon = \frac{8\pi V}{c^3 h^3} \epsilon^2 d\epsilon$$

$$\rightarrow \ln \mathcal{Z} = - \int d\epsilon \frac{8\pi V}{c^3 h^3} \epsilon^2 \ln(1 - e^{-\beta \epsilon})$$

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$$\ln \tilde{Z} = - \int_0^\infty d\epsilon \left(\frac{8\pi V}{c^3 h^3} \right) \epsilon^2 \ln(1 - e^{-\beta \epsilon}) = - \frac{8\pi V}{c^3 h^3} I(\beta)$$

integration by parts

$$I(\beta) = - \int_0^\infty d\epsilon \frac{\epsilon^3}{3} \frac{1}{1 - e^{-\beta \epsilon}} \stackrel{x = \beta \epsilon}{=} - \frac{\beta^3}{3} \int_0^\infty dx \frac{x^3}{e^x - 1} = - \frac{\beta^3}{3} \frac{\pi^4}{15}$$

$$\int_0^\infty dx \frac{x^3}{e^x - 1} = \sum_{n=0}^\infty \int_0^\infty dx e^{-(n+1)x} x^3 = \sum_{n=0}^\infty \frac{6}{(n+1)^4}$$

$$\Rightarrow \ln \tilde{Z} = \frac{8\pi^5 V}{15 c^3 h^3} \beta^{-3} \Rightarrow J = -kT \ln \tilde{Z} = -PV \langle E \rangle - \frac{\partial \ln \tilde{Z}}{\partial \beta}$$

$$\langle E \rangle = - \frac{\partial \ln \tilde{Z}}{\partial \beta} = - \frac{8\pi^5 V}{15 c^3 h^3} (-3) \beta^{-4} = \frac{8\pi^5}{15 c^3 h^3} \beta^{-4} V = \langle E \rangle$$

$$\Rightarrow \langle E \rangle = \left(\frac{\langle E \rangle}{V} \right) = \frac{8\pi^5}{15 c^3 h^3} (kT)^4 \propto T^4$$

related to the dimensionality of spacetime here

$$J = -P \cdot V \Rightarrow PV = \frac{1}{3} \langle E \rangle V \Rightarrow P = \frac{1}{3} \langle E \rangle$$

total average energy density

$$S = \frac{\langle E \rangle - J}{T} = \frac{\langle E \rangle - (-PV)}{T} = \frac{\langle E \rangle + PV}{T} = \frac{4}{3} \frac{\langle E \rangle V}{T}$$

$$\Rightarrow S = 4 \cdot \frac{PV}{T} = \frac{4}{3} V \frac{\langle E \rangle}{T} \Rightarrow S \propto \frac{4}{3} VT^3$$

What will happen if the radiation undergoes a reversible adiabatic change

$$VT^3 = \text{const} \quad PV^{1/3} = \text{const}$$

$$PV^\gamma = \text{const, classical gas} \Rightarrow \gamma \sim \frac{C_p}{C_v}, \text{ Not true in QM}$$

$$C_v = T \left(\frac{\partial S}{\partial T} \right)_V = 3S \sim T^3, \quad \frac{C_p}{C_v} = \frac{\left(\frac{\partial V}{\partial P} \right)_T}{\left(\frac{\partial V}{\partial P} \right)_S} = \frac{1}{\left(\frac{\partial P}{\partial V} \right)_T} \cdot \frac{1}{\left(\frac{\partial P}{\partial V} \right)_S} \rightarrow \infty$$

Recap: "Black body" $\leftrightarrow$ gas of photons, bosons, c

1) Relativistic, $c \rightarrow U_c(1, \epsilon)$

2) N not conserved, $M=0$

grand canonical $\sim$ canonical

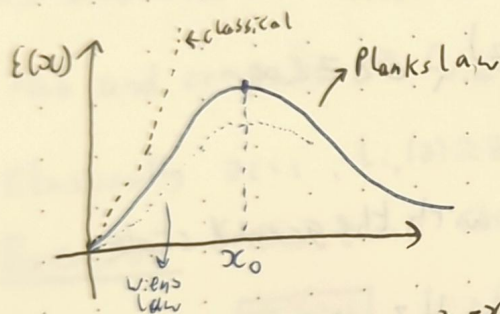
$$\langle E \rangle = \frac{\langle E \rangle}{V} = \frac{8\pi^5}{15 c^3 h^3} \beta^{-4}$$

We can define, since $\langle \epsilon \rangle = \int_0^\infty d\epsilon \frac{W_c(l, \epsilon)}{V} \epsilon \langle n(\epsilon) \rangle$

$$\epsilon(\epsilon) \stackrel{de}{=} \frac{W_c(l, \epsilon)}{V} \epsilon \langle n(\epsilon) \rangle_{\mu=0} d\epsilon$$

express in terms of frequency ω using $\omega = \frac{\epsilon}{\hbar}$

$$\Rightarrow \epsilon(\omega) d\omega = \frac{\hbar \omega^3}{\pi^2 c^3} \frac{1}{e^{\beta \hbar \omega} - 1} d\omega \Rightarrow \epsilon(x) dx \propto \frac{x^3}{e^x - 1} dx$$



$$x_0 = \beta_0 \hbar \omega_0 \Rightarrow \omega_0 = \hbar \frac{1}{\beta_0} \frac{k T_0}{\hbar}$$

This determines the max of the distribution determined by $\frac{d\epsilon(x)}{dx} \Big|_{x_0} = 0$

$$QM \quad x \gg 1 \rightarrow \epsilon(x) \approx x^3 e^{-x}$$

$$CM \quad x \ll 1 \rightarrow \epsilon(x) \approx x^2 \rightarrow \text{Ultraviolet Catastrophe!}$$

Stat Phys 2 NR Fermionic Gas

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 $0 < z < \infty$

$$\ln \tilde{Z} = \sum_i \ln(1 + z e^{-\beta \epsilon_i}) = \ln(1+z) + \int d\epsilon W(\epsilon) \ln(1 + z e^{-\beta \epsilon})$$

$$W(\epsilon) d\epsilon = (\deg) \frac{V}{h^3} (2m)^{3/2} 2\pi \epsilon^{1/2} d\epsilon$$

$$\Rightarrow \ln \tilde{Z} = \ln(1+z) - (\dots) \int_0^{\infty} \frac{dx x^{3/2}}{1 + \frac{e^x}{z}} \quad \text{Li}_{5/2}(-z)$$

$$Li_r(-z) = -\frac{1}{\Gamma(r)} \int_0^{\infty} \frac{dt t^{r-1}}{1 + \frac{e^t}{z}}$$

↑
converges for $z \geq -1$

$$\rightarrow \ln \tilde{Z} = \ln(1+z) - (\deg) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-3/2} Li_{5/2}(-z), \quad 0 < z < \infty$$

• there is no bound on z & hence on $Li_{5/2}$
 thus there is no special physics associated with the ground state

• The polylog term dominates for both large & small $z \Rightarrow$ we can neglect the ground state contribution

$$\rightarrow \ln \tilde{Z} = -(\deg) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-3/2} Li_{5/2}(-z)$$

$$\langle N \rangle = z \frac{\partial \ln \tilde{Z}}{\partial z} = -(\deg) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-3/2} Li_{3/2}(-z) \quad \text{A.M.W.}$$

$$Li_r(-z) \underset{z \ll 1}{\approx} -z \left\{ 1 - \frac{z}{2} + \dots \right\}, \quad Li_r(-z) \underset{z \gg 1}{\approx} -\frac{(\ln z)^r}{\Gamma(r+1)} \left\{ 1 + \frac{\pi^2}{6} \frac{\Gamma(r-1)}{(\ln z)^2} \right\}$$

$$\langle E \rangle = -\frac{\partial \ln \tilde{Z}}{\partial \beta} \Big|_{V,N} = -\frac{3}{2} (\deg) \left(\frac{2\pi m}{h^2} \right)^{3/2} V \beta^{-5/2} Li_{5/2}(-z) = \frac{3}{2} P \cdot V$$

$$\text{but also } \langle E \rangle = \frac{3}{2} P \langle N \rangle \frac{Li_{5/2}(-z)}{Li_{3/2}(-z)} = \begin{cases} z \gg 1 \rightarrow \frac{\langle N \rangle}{\beta} \ln z & \text{"QM"} \\ z \ll 1 \rightarrow \frac{3}{2} kT \langle N \rangle & \text{"classical"} \end{cases}$$

$$F = \langle E \rangle - T S = \mu \langle N \rangle + J$$

$$\ln z = \beta \mu \rightarrow \mu = \langle N \rangle kT \left\{ \ln z - \frac{Li_{5/2}(-z)}{Li_{3/2}(-z)} \right\}$$

$$S = \frac{1}{T} (\langle E \rangle - F) = \langle N \rangle k \left\{ \frac{5}{2} \frac{Li_{5/2}(-z)}{Li_{3/2}(-z)} - \ln z \right\}$$

$$C_{V,N} = T \left(\frac{\partial S}{\partial T} \right)_{V,N} = T \left(\frac{\partial S}{\partial z} \right)_{V,N} \cdot \left(\frac{\partial z}{\partial T} \right)_{V,N}$$

$$\left(\frac{\partial z}{\partial T} \right)_{V,N} = ?$$

Stat Phys 2 N-R Fermionic Gas

28/02

$$\left(\frac{\partial Z}{\partial T}\right)_{V,N} = ? \quad \text{Set } N = \text{fixed} = \langle N \rangle \Rightarrow \left(\frac{\partial N}{\partial T}\right)_{V,N} = 0$$

$$\Rightarrow 0 = (\text{deg}) V \left(\frac{2\pi m k}{h^2}\right)^{3/2} \left\{ -\frac{3}{2} T^{1/2} \text{Li}_{5/2}(-z) - T \frac{\partial}{\partial T} \text{Li}_{5/2}(-z) \right\}$$

$$\Rightarrow -\frac{3}{2} T^{1/2} \text{Li}_{5/2}(-z) = -T^{3/2} \frac{\text{Li}_{5/2}(-z)}{(-z)} \left(\frac{\partial z}{\partial T}\right)_{V,N}$$

$$\Rightarrow \frac{1}{z} \left(\frac{\partial z}{\partial T}\right)_{V,N} = -\frac{3}{2T} \frac{\text{Li}_{5/2}(-z)}{\text{Li}_{3/2}(-z)}$$

$$\text{sub and compute} \rightarrow C_{V,N} = -\frac{9}{4} N k \frac{\text{Li}_{5/2}(-z)}{\text{Li}_{3/2}(-z)} + \frac{15}{4} N k \frac{\text{Li}_{5/2}(-z)}{\text{Li}_{3/2}(-z)}$$

$$\text{classically } z \ll 1, \text{Li}_\nu(z) \approx z \rightarrow C_{V,N} \approx \frac{3}{2} N k$$

Exercise Show that classical $\rightarrow S$ behaves as expected

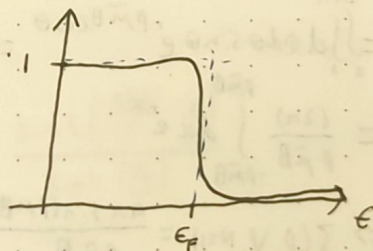
Not doing exercise $\boxed{z \gg 1} \quad \text{Li}_{5/2}(-z) \underset{z \gg 1}{\approx} -\frac{(\ln z)^{3/2}}{\Gamma(5/2)} = -\frac{\langle N \rangle}{V} \left[\left(\frac{h^2}{2\pi m}\right)^{3/2} \frac{1}{(\text{deg})} \right] \beta^{3/2}$
solving in terms of $\langle N \rangle$

$$\Rightarrow \mu(z \gg 1) \approx \# \left(\frac{N}{V}\right)^{2/3} \equiv \text{fixed independent of } T$$

μ is a constant at small temperatures

$$\text{As } T \rightarrow 0 \quad u(T \rightarrow 0) \approx E_F \quad \text{fermi energy}$$

$$\rightarrow \langle n(E) \rangle = \frac{1}{e^{\beta(E-\mu)} + 1} \underset{T \rightarrow 0}{=} \begin{cases} 1 & E < E_F \\ 0 & E > E_F \end{cases}$$



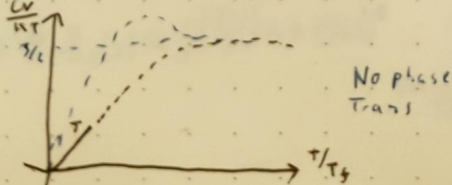
$$\begin{aligned} N &= \int_0^\infty dE W_c(1, E) \langle n(E) \rangle \underset{T \gg 1}{=} \int_0^{E_F} dE W_c(1, E) = \# E_F^{3/2} \\ \langle E \rangle &\underset{T \gg 0}{=} \int_0^\infty dE E \langle n(E) \rangle W_c(1, E) = \int_0^{E_F} dE E W_c(1, E) = \# E_F^{5/2} \end{aligned} \quad \left. \vphantom{\int_0^\infty} \right\} \Rightarrow \frac{3}{5} N E_F = \langle E \rangle$$

let's study C_V for $\beta \gg 1$ ($z \gg 1$)

Subbing in leading polylog terms sends C_V to 0 not good
 Keep more terms in polylog expansion

$$\text{Doing so} \Rightarrow C_V \underset{T \rightarrow 0}{\approx} \frac{\pi^2}{2} N k \left(\frac{kT}{E_F}\right) \propto T$$

unlike bosons which are T^3 , so we can tell what we have with this behaviour
 $T_3 \sim \frac{E_F}{k}$

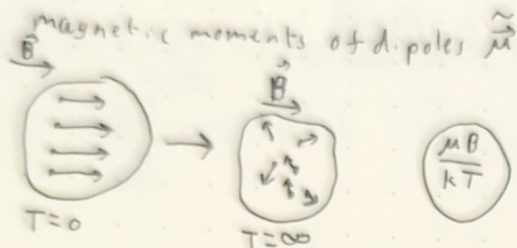


Statphys 2 Statistics of Paramagnetism

13/03

N magnetic dipoles interacting with a magnetic $\vec{B}$

Look classical first
but the exam will be
focus on quantum



Completely localised dipoles

Classical Treatment

assume all dipoles
have $|\vec{\mu}| = \mu$

$$E = \sum_{i=1}^N E_i = - \sum_{i=1}^N \vec{\mu}_i \cdot \vec{B} = - \sum_{i=1}^N \mu B \cos \theta_i$$

Thermo limit: $Z = \prod_{i=1}^N \mathcal{Z}_i = \mathcal{Z}^N$ because there is no interaction between the dipoles

$$\mathcal{Z} = \mathcal{Z}(T, V, 1) = \int_{\Omega} \mu B \cos \theta e^{-\beta \mu B \cos \theta} d\Omega$$

not really a sum, θ, ϕ continuous

$$= \int_0^{2\pi} d\phi \int_0^\pi \sin \theta e^{-\beta \mu B \cos \theta} d\theta = 2\pi \int_{-1}^1 d(\cos \theta) e^{+\beta \mu B \cos \theta}$$

$$x = \beta \mu B \cos \theta$$

$$\rightarrow = \frac{(2\pi)}{\beta \mu B} \int_{-\beta \mu B}^{+\beta \mu B} dx e^{+x}$$

$$\Rightarrow \mathcal{Z}(\beta, V, N=1) = \frac{4\pi \mu B \sinh(\beta \mu B)}{\beta \mu B} \Rightarrow Z = \mathcal{Z}^N = \left(\frac{4\pi \mu B \sinh(\beta \mu B)}{\beta \mu B} \right)^N$$

$$F = -kTN \ln \mathcal{Z}$$

$$\langle \vec{\mu} \cos \theta \rangle \equiv \text{average dipole moment of each particle} = \langle M \rangle$$

$$N \langle \vec{\mu} \cos \theta \rangle \equiv \text{all of the system}$$

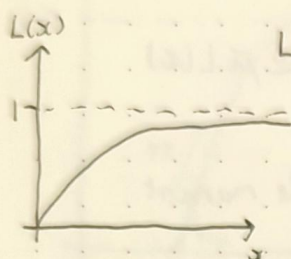
$$\langle \mu \cos \theta \rangle = \frac{\int \int d\phi d\theta \sin \theta \mu \cos \theta e^{-\beta \mu B \cos \theta}}{\int \int d\phi d\theta \sin \theta e^{-\beta \mu B \cos \theta}} = \frac{1}{\mathcal{Z}} \left(\frac{\partial \mathcal{Z}}{\partial \beta} \right) = \frac{1}{\mathcal{Z}} \left(\frac{\partial \ln \mathcal{Z}}{\partial \beta} \right)_{V, N}$$

$$\Rightarrow -\frac{1}{N} \left(\frac{\partial F}{\partial B} \right) = \langle M \rangle$$

$$\chi_T \equiv \text{isothermal mag. susceptibility} = \left(\frac{\partial \langle M \rangle}{\partial B} \right)_{\beta, V, N}$$

So let us evaluate: $\langle M \rangle = \dots = -\frac{1}{\beta B} + \tilde{M} \coth(\beta \tilde{M} B)$

$$= \tilde{M} \left(\underbrace{-\frac{1}{\beta \tilde{M} B}}_{L(x)} + \coth(\beta \tilde{M} B) \right)$$



$$L(x) = \coth(x) - \frac{1}{x}$$

$$L(x) \quad x \equiv \beta \tilde{M} B$$

1) $x \gg 1, \tilde{M} B \gg kT$

$$L(x) \rightarrow 1$$

$$\langle M \rangle = \tilde{M}$$

2) $x \ll 1, \tilde{M} B \ll kT$

$$L(x) \approx \frac{x}{3} - \frac{x^3}{45} + \mathcal{O}(x^5)$$

$$\langle M \rangle \approx \frac{M B}{3 k T} \rightarrow \langle M \rangle \propto \frac{1}{T} \quad \leftarrow \text{Curie law paramag}$$

$$\hat{H} = -\vec{\mu} \cdot \vec{B} \quad \text{quantisation}$$

$$\vec{\mu} = g \left(\frac{q}{2mc} \right) \cdot \left\{ \begin{array}{l} \uparrow \\ \text{combination } \vec{J} \end{array} \right.$$

↑
gyromagnetic ratio

$$\vec{\mu} = -g \tilde{\mu}_0 \cdot \vec{m} \cdot \vec{B}$$

↑
quantisation no $m = -J, -J+1, \dots, J$

$$\tilde{\mu}_0 = \frac{q \hbar}{2mc}$$

$$\mathcal{Z}(\beta, V, B, N=1) = \sum_{m=-J}^J e^{\beta g \tilde{\mu}_0 B m}$$

$$\Rightarrow \mathcal{Z} = \sum_{m=-J}^J e^{-\frac{m x}{J}} = e^{-x} \left(\frac{e^{\frac{2J+1}{J} x} - 1}{e^{\frac{x}{J}} - 1} \right) = \frac{\sinh\left(\frac{2J+1}{J} x\right)}{\sinh\left(\frac{x}{J}\right)}$$

$x = \beta g \mu_0 B J$

$$Z = \mathcal{Z}, F = -kT \ln Z \rightarrow \langle M \rangle = \frac{1}{\beta} \left(\frac{\partial \ln \mathcal{Z}}{\partial B} \right)_{\beta, V}, \chi_T = \left(\frac{\partial \langle M \rangle}{\partial B} \right)$$

$$\langle M \rangle = \frac{1}{\beta} \left(\frac{\partial \ln \mathcal{Z}}{\partial B} \right)_{\beta, V} = g \mu_0 J \left\{ \frac{2J+1}{2J} \coth \left[x \left(\frac{2J+1}{2J} \right) \right] - \frac{1}{2J} \coth \left(\frac{x}{2J} \right) \right\}$$

$\mathcal{B}_J(x)$: Brillouin function order J

① $x \gg 1$ (Strong magnetic vs Temp)

$$\mathcal{B}_J(x) \approx 1 \quad \forall J$$

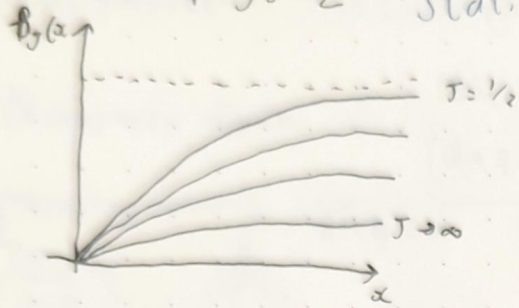
magnetic saturation

② $x \ll 1$ (weak magnetic vs Temp)

$$\mathcal{B}_J(x) \approx \frac{1}{J} \left(\frac{J+1}{J} \right) x + \mathcal{O}(x^3) \Rightarrow \langle M \rangle \approx \frac{g^2 \tilde{\mu}_0^2 J(J+1) H}{3k}$$

only leading term

Stat Phys 2 Statistics of Paramagnets 14/03



When do we get the classical Result?

$$T \rightarrow \infty, g \rightarrow 0 \quad (\epsilon_m = g \mu_B \tilde{\mu}_0 B)$$

$$\left. \begin{aligned} &B_z(x) \text{ indep of } T \\ &\coth\left(\frac{x}{2T}\right) \approx \frac{2T}{x} \end{aligned} \right\} \rightarrow \langle M \rangle \approx \tilde{\mu}_0 L(x)$$

$\rightarrow g \tilde{\mu}_0 \equiv \tilde{\mu}$ classical dipole moment

Now check when we fix $T = \frac{1}{2}$

$$\Rightarrow \epsilon_i = \begin{cases} -\tilde{\mu}_0 B \equiv -\epsilon_0 \\ +\tilde{\mu}_0 B \equiv \epsilon_0 \end{cases}$$

Thermal limit, canonical ensemble

$$Z = \sum \delta^N = \left(\frac{2 \cosh(2\epsilon_0)}{2} \right)^N = 2^N \cosh^N(x), \quad x = \beta \tilde{\mu}_0 B = \beta \epsilon_0$$

$$F = -kT \ln Z = -kTN \ln [2 \cosh(\beta \epsilon_0)]$$

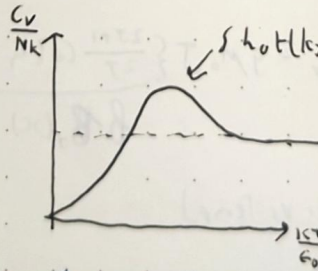
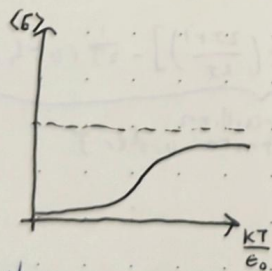
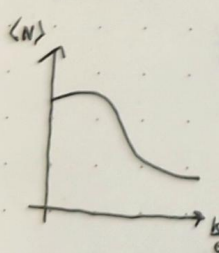
$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,B,N} = Nk \{ \ln [2 \cosh(\beta \epsilon_0)] - \beta \epsilon_0 \tanh(\beta \epsilon_0) \}$$

$$\langle E \rangle = F + TS = -\frac{\partial \ln Z}{\partial \beta} = -N \epsilon_0 \tanh(\beta \epsilon_0) = -N \langle M \rangle B$$

$$\langle M \rangle = -\frac{1}{N} \left(\frac{\partial F}{\partial B} \right)_{T,V,N} = \tilde{\mu}_0 \tanh(\beta \epsilon_0) \quad C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,B,N} = \frac{Nk(\beta \epsilon_0)^2}{\cosh^2(\beta \epsilon_0)}$$

$$S \approx Nk [\ln(2) - (\beta \epsilon_0)^2] \approx Nk \ln 2 \quad \Rightarrow \Omega = 2^N, kT \gg \epsilon_0 \text{ disorder}$$

$$S \approx Nk [\ln 2 - \beta \epsilon_0] \approx 0 \quad \Rightarrow \Omega = 0, kT \ll \epsilon_0 \text{ order all in line with } B \text{ field}$$



Why can't they all be ordered opposite to B 's field

Solve for T from $\langle E \rangle \Rightarrow \frac{1}{T} = \frac{k}{2\epsilon_0} \ln \left(\frac{1 - \frac{\langle E \rangle}{N\epsilon_0}}{1 + \frac{\langle E \rangle}{N\epsilon_0}} \right)$

$\Rightarrow T < 0$ is a characteristic feature of systems with upper bound on energy

$$S = N k \ln 2$$

$\langle E \rangle \Rightarrow 0$

↖ half positive
half negative

- when $\langle E \rangle \rightarrow 0^+$ what happens

$$\frac{\partial S}{\partial E} \sim \frac{1}{T} < 0$$

In this case $\exists T$ s.t. $kT \gg \frac{E}{N}$

Canonical Ensemble: $Z \approx (\sum_n e^{-\beta E_n})^N$ neglecting interactions for high T

$$\beta E_n \ll 1 \Rightarrow Z \simeq \left(\sum_n \left(1 - \beta E_n + \frac{1}{2} \beta^2 E_n^2 \right) \right)^N$$

$$\ln Z \approx N \ln \left[\sum_n 1 - \sum_n \rho \epsilon_n + \frac{1}{2} \sum_n \rho^2 \epsilon_n^2 \right]$$

To proceed we set $\sum_n 1 \approx g$, e.g. all possible orientations in a magnetic field,
all possible polarizations etc.

$$\sum p_n \epsilon_n \approx g\beta \langle \epsilon_n \rangle$$

$$\sum_n \beta^2 \epsilon_n^2 \approx g \beta^2 \langle \epsilon_n^2 \rangle$$

$$\text{Sub into } \ln Z \Rightarrow \ln Z \simeq N \ln \left[g(1 - \rho \langle \epsilon_n \rangle + \frac{1}{2} \rho^2 \langle \epsilon_n^2 \rangle) \right]$$

$$\overline{P(\epsilon_n < \epsilon_1)} N \left\{ \ln g - \left[P(\epsilon_n) - \frac{1}{2} P^2(\epsilon_n) \right] - \frac{1}{2} \left[P(\epsilon_n) - \frac{1}{2} P^2(\epsilon_n) \right]^2 + \dots \right\}$$

$$\ln Z \simeq N [\ln g - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 (\langle \epsilon_n^2 \rangle - \langle \epsilon_n \rangle^2)]$$

$$F = -kT \ln Z \simeq -NkT \ln g + N \langle E_n \rangle - \frac{N}{2kT} \langle (E - \langle E_n \rangle)^2 \rangle$$

$$S \simeq N k \ln g - \frac{Nk}{2} \beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad \langle E \rangle \simeq N \langle \epsilon_n \rangle - N \beta \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle$$

$C_v \approx Nk\beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \geq 0$; These formulae determine the thermobehavior of the

$$\frac{L}{N K}$$

← valid around system, for $\beta \rightarrow 0^+$ or $\beta \rightarrow 0^-$
here



8.

Exercise

ideal gas in constant $\vec{B}$ enclosed in a volume V
 find $\uparrow$ average dipole moment $\uparrow$ Strong/Weak $\vec{B}$ fields
 $\uparrow$ magnetic susceptibility $\uparrow$ Large/Small T , Curie law?

• ideal gas of indistinguishable spin- $\frac{1}{2}$ particles in $\vec{B}$

$$\ln \tilde{Z} = - \sum_n \ln(1 + e^{-\beta \epsilon_n} e^{\beta \mu B}) = - \sum_n \ln(1 + e^{-\beta \epsilon_n(P) \pm \beta \tilde{\mu} B} e^{\beta \mu B})$$

$$= \sum_n \ln(1 + e^{-\beta \epsilon_n(P)} e^{\beta \tilde{\mu} B}) + \sum_n \ln(1 + e^{-\beta \epsilon_n(P)} e^{\beta \mu B})$$

$$\mu_{\pm} = \mu \pm \tilde{\mu} B$$

$$z_{\pm} = e^{\beta \mu_{\pm}}$$

$$\ln \tilde{Z} = \sum_n \ln(1 + z_+ e^{-\beta \epsilon_n(P)}) + \sum_n \ln(1 + z_- e^{-\beta \epsilon_n(P)})$$

prev section

$$\Rightarrow J = -kT \ln \tilde{Z} \simeq (\text{deg}) \frac{kTV}{\lambda_T^3} (L_{i_{5/2}}(z_+) + L_{i_{5/2}}(z_-))$$

$$\lambda_T = \left(\frac{h^2}{2\pi m kT} \right)^{1/2}$$

$$\langle M \rangle = -\frac{1}{N} \left(\frac{\partial J}{\partial B} \right)_{T, \mu, V}, \quad \chi_T = \left(\frac{\partial \langle M \rangle}{\partial B} \right)$$

N_+ , N_- - Number that are parallel and antiparallel to $\vec{B}$

$$\langle M \rangle = \frac{\tilde{\mu}}{N} (N_+ - N_-)$$

for small temps: $\lambda_T = \frac{\partial \langle M \rangle}{\partial B} = -\frac{V}{N} \frac{\tilde{\mu}}{\lambda_T^3} (\beta \tilde{\mu}) [L_{i_{5/2}}(z_+) + L_{i_{5/2}}(z_-)]$

$$\simeq \frac{-2\beta \tilde{\mu}^2 V}{\lambda_T^3} L_{i_{5/2}}(-z) \Rightarrow \text{show Curie Law is valid.}$$

Black Body

Paul Haskins

March 10, 2025

Black body implies a gas of phonons, bosonic particles.

1. Relativistic $\omega_c(1, \epsilon)$ 2. N not conserved, $\mu = 0$ grand canonical is approx canonical and J approx is F .
The total average energy density is given by

$$\langle \epsilon \rangle = \frac{\langle E \rangle}{V} = \frac{8\pi^5}{15c^3 h^3} \beta^{-4} \quad (1)$$

$$\langle \epsilon \rangle = \int_0^\infty d\epsilon \frac{\omega_c(1, \epsilon)}{V} \epsilon \langle n(\epsilon) \rangle \quad (2)$$

We could define

$$E(\epsilon) d\epsilon = \frac{\omega_c}{V} \epsilon \langle n(\epsilon) \rangle_{\mu=0} d\epsilon \quad (3)$$

Above is the average energy distribution of photons.

Let us express it in terms of the frequency ω . Using $\epsilon = \hbar\omega$ we have

$$E(\omega) d\omega = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta\hbar\omega} - 1} \quad (4)$$

$$\implies_{x=\beta\hbar\omega} E(x) \propto \frac{x^3}{e^x - 1} \quad (5)$$

$$x_0 = \beta_0 \hbar \omega_0$$

$\implies \omega_0 = \frac{kT_0}{\hbar}$. This determines the max of the distribution determined by

$$\left. \frac{dE(x)}{dx} \right|_{x_0} = 0 \quad (6)$$

We can examine limits.

$E(x) \approx x^3 e^x$ for $x \gg 1$. QM

$E(x) \approx x^2$ for $x \ll 1$ classical.

This is known as Wien's law. UV-catastrophe.

The only difference between photons and phonons is the factor of the degeneracy where photons have a deg of 2 while phonons have 3N as they have mass.

Non-Relativistic fermionic gas

$$\ln(\Xi) = \sum_{n=1}^{\infty} \ln(1 + ze^{-\beta\epsilon_n}) \quad (7)$$

This holds for $0 < z < \infty$

$$= \ln(1+z) + \int_0^\infty d\epsilon \omega_c \ln(1 + ze^{-\beta\epsilon}) \quad (8)$$

Previously we had

$$\omega_c d\epsilon = (deg) \frac{V}{h^3} (2m)^{\frac{3}{2}} 2\pi \epsilon^{\frac{1}{2}} d\epsilon \quad (9)$$

Using IBP and letting $x = \beta\epsilon$

$$\ln(\Xi) = \ln(1+z) - (\dots) \int_0^\infty \frac{dx x^{\frac{3}{2}}}{1 + \frac{e^x}{z}} \quad (10)$$

Using analytical cont for $-z$ and the def of polylogarithm.

Our integral is $\mathcal{L}_{i\frac{5}{2}}(-z)$

$$\mathcal{L}_{i\frac{5}{2}}(-z) = -\frac{1}{\Gamma(r)} \int_0^\infty \frac{dt t^{(r-1)}}{1 + \frac{e^t}{z}} \quad (11)$$

1.png

Figure 1: Planck's law. The monotonic dashed line is the classical case.

which converges for $z \geq -1$

$$\ln \Xi = -\ln(1+z)(deg) \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{3}{2}} \mathcal{L}_{i\frac{5}{2}}(-z) \quad (12)$$

holds for $0 \leq z \leq \infty$

- * There is no bound on z and hence on $\mathcal{L}_{i\frac{5}{2}}$. Thus there is no special physics associated with the ground state.
- * * The polylogarithm term dominates for both large and small z and combining with * we can safely neglect the ground state contribution.

$$\ln \Xi = -(deg) \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{3}{2}} \mathcal{L}_{i\frac{5}{2}}(-z) \quad (13)$$

$$\langle N \rangle = z \frac{d \ln \Xi}{dz} = (deg) \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{3}{2}} \mathcal{L}_{i\frac{3}{2}}(-z) \quad (14)$$

$$\langle E \rangle = -\frac{d \ln \Xi}{d\beta} \Big|_{v,z} = -\frac{3}{2} (deg) \left(\frac{2\pi m}{h^2} \right)^{\frac{3}{2}} V \beta^{-\frac{5}{2}} \mathcal{L}_{i\frac{3}{2}}(-z) = \frac{3}{2} PV$$

(16)

$$\mathcal{L}_{i\frac{5}{2}}(-z) \approx -z \left[1 - \frac{z}{2^r} + \dots \right] \quad (17)$$

for $z \ll_1$ and for $z \gg 1$.

$$\mathcal{L}_{i\frac{5}{2}}(-z) \approx -\frac{(\ln(z))^r}{\Gamma(r+1)} \left[1 + \frac{\pi^2 r(r-1)}{(\ln(z))^2} \right] \quad (18)$$

But also

$$\langle E \rangle = \frac{3}{2\beta} \langle N \rangle \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} = \begin{cases} \frac{\langle N \rangle}{\beta} \ln(z) & \text{if } z \gg_1 \\ \frac{3}{2} kT \langle N \rangle & \text{if } z \ll_1 \end{cases} \quad (20)$$

$$F = \langle E \rangle - TS = \mu \langle N \rangle + J = \langle N \rangle kT \left[\ln(z) - \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} \right] \quad (21)$$

where in the last equality $\beta\mu = \ln(z)$

$$S = \frac{1}{T} (\langle E \rangle - F) = \langle N \rangle k \left[\frac{5}{2} \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} \right] \quad (22)$$

$$C_{v,n} = T \frac{\partial S}{\partial T} |_{V,N} = T \frac{\partial S}{\partial z} |_{V,N} \frac{\partial z}{\partial T} |_{V,N} \quad (23)$$

what is $\frac{\partial z}{\partial T}$

Set N fixed $= \langle N \rangle \implies \frac{\partial N}{\partial T} |_{V,N} = 0$

$$0 = (deg) V \left(\frac{2\pi m k}{h^2} \right)^{\frac{3}{2}} \left[-\frac{3}{2} T^{\frac{1}{2}} \mathcal{L}_{i\frac{3}{2}} - T^{\frac{3}{2}} \frac{\partial}{\partial T} \mathcal{L}_{i\frac{5}{2}} \right] \quad (24)$$

$$-\frac{3}{2} T^{\frac{1}{2}} \mathcal{L}_{i\frac{3}{2}} = -T^{\frac{3}{2}} \frac{\mathcal{L}_{i\frac{5}{2}}}{-z} \frac{\partial z}{\partial T} \quad (25)$$

$$\frac{1}{z} \frac{\partial z}{\partial T} = -\frac{3}{2T} \frac{\mathcal{L}_{i\frac{3}{2}}(-z)}{\mathcal{L}_{i\frac{1}{2}}(-z)} \quad (26)$$

Sub and compute.

$$C_{V,N} = -\frac{9}{4} N k \frac{\mathcal{L}_{i\frac{3}{2}}(-z)}{\mathcal{L}_{i\frac{1}{2}}(-z)} + \frac{15}{4} N k \frac{\mathcal{L}_{i\frac{5}{2}}(-z)}{\mathcal{L}_{i\frac{3}{2}}(-z)} \quad (27)$$

Classically $C_{V,N} \approx \frac{3}{2} N k$ as expected the leading behavior of $\mathcal{L}_{ir}$ for $z \ll 1$.

Exercises: Show that in the canonical regime S behaves as expected.

$$\mathcal{L}_{i\frac{3}{2}}(-z) \approx -\frac{(\ln(z))^{\frac{3}{2}}}{\Gamma(\frac{5}{2})} = -\frac{\langle N \rangle}{V} \left[\left(\frac{h^2}{2\pi m} \right)^{\frac{3}{2}} \beta^{\frac{3}{2}} \frac{1}{(deg)} \right] \quad (28)$$

In last equals we solve in terms of $\langle N \rangle$.

$$\implies \mu(z \gg_1) \approx \left(\frac{N}{V} \right)^{\frac{2}{3}} \quad (29)$$

which is fixed and independent of T .

As $T \rightarrow 0$, $\mu(T \rightarrow 0) \approx \exists_F$

What does this mean for $\langle n(\epsilon) \rangle$?
Under the limit $T \rightarrow 0$

$$\langle n(\epsilon) \rangle = \frac{1}{e^{\beta(\epsilon - \mu)} + 1} = \begin{cases} 1 & \text{if } \epsilon < \epsilon_f \\ 0 & \text{if } \epsilon > \epsilon_F \end{cases}$$

(31)

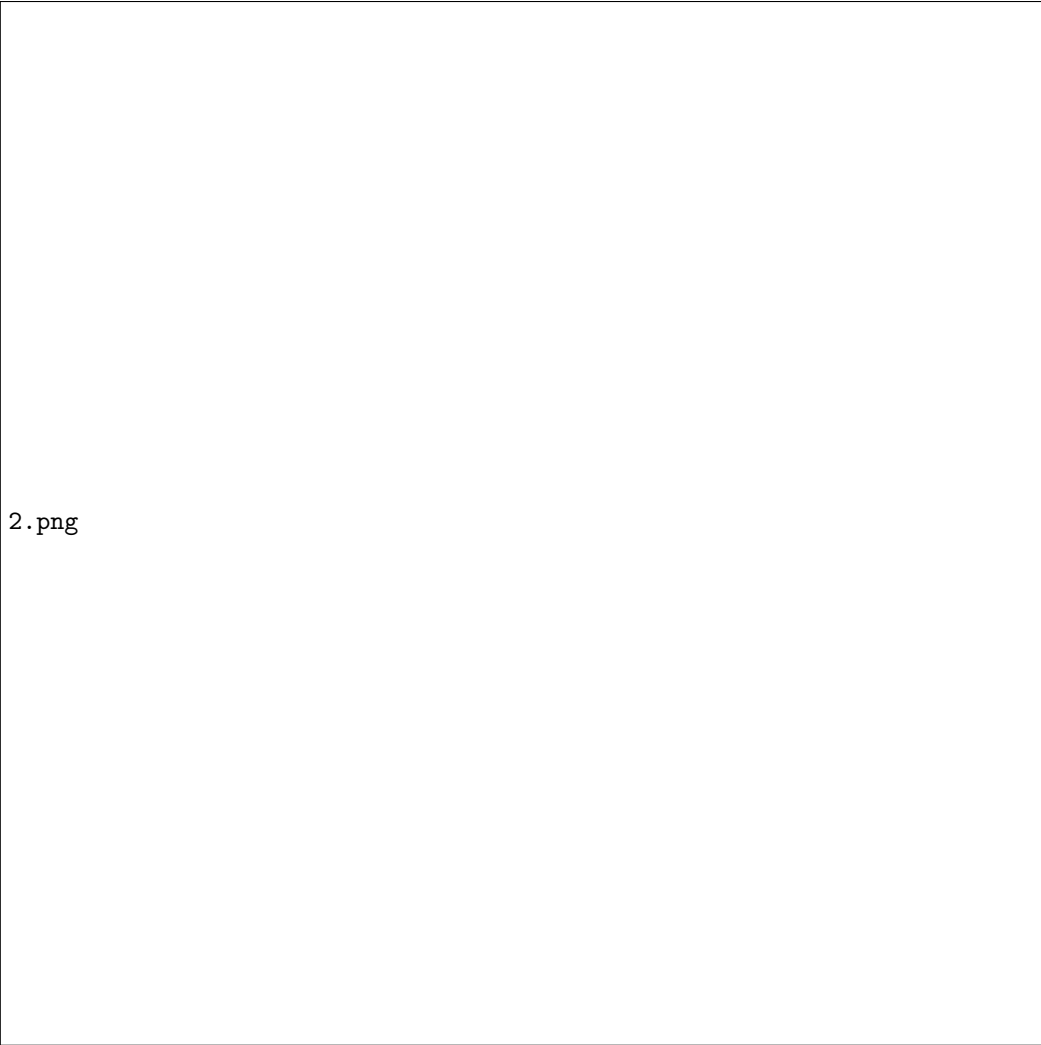


Figure 2: 2.png

$$N = \int_0^\infty d\epsilon \omega_c \langle n(\epsilon) \rangle \quad (32)$$

As $T \rightarrow 0$ with $z \gg 1$.

$$\Rightarrow \int_0^{\epsilon_F} d\epsilon \omega_c = \epsilon_F^{\frac{3}{2}} \quad (33)$$

$$\langle E \rangle = \int_0^\infty d\epsilon \epsilon \langle n(\epsilon) \rangle = \int_0^{\epsilon_F} d\epsilon \epsilon \omega_c = \epsilon_F^{\frac{5}{2}} \quad (34)$$

$$\Rightarrow \langle E \rangle = \frac{3}{5} N \epsilon_F \quad (35)$$

Now let us study C_v for $\beta \gg 1$ ($z \gg 1$) Substituting the leading polylogarithms behavior in C_v results to zero!
We need to keep higher order terms in the second expansion in the box ??

Doing so

$$\Rightarrow C_v \approx \frac{\pi^2}{2} N k \left(\frac{kT}{\epsilon_F} \right) \quad (36)$$

we can tell what system (bosonic or fermionic) we are dealing with by the small T behavior of C_v

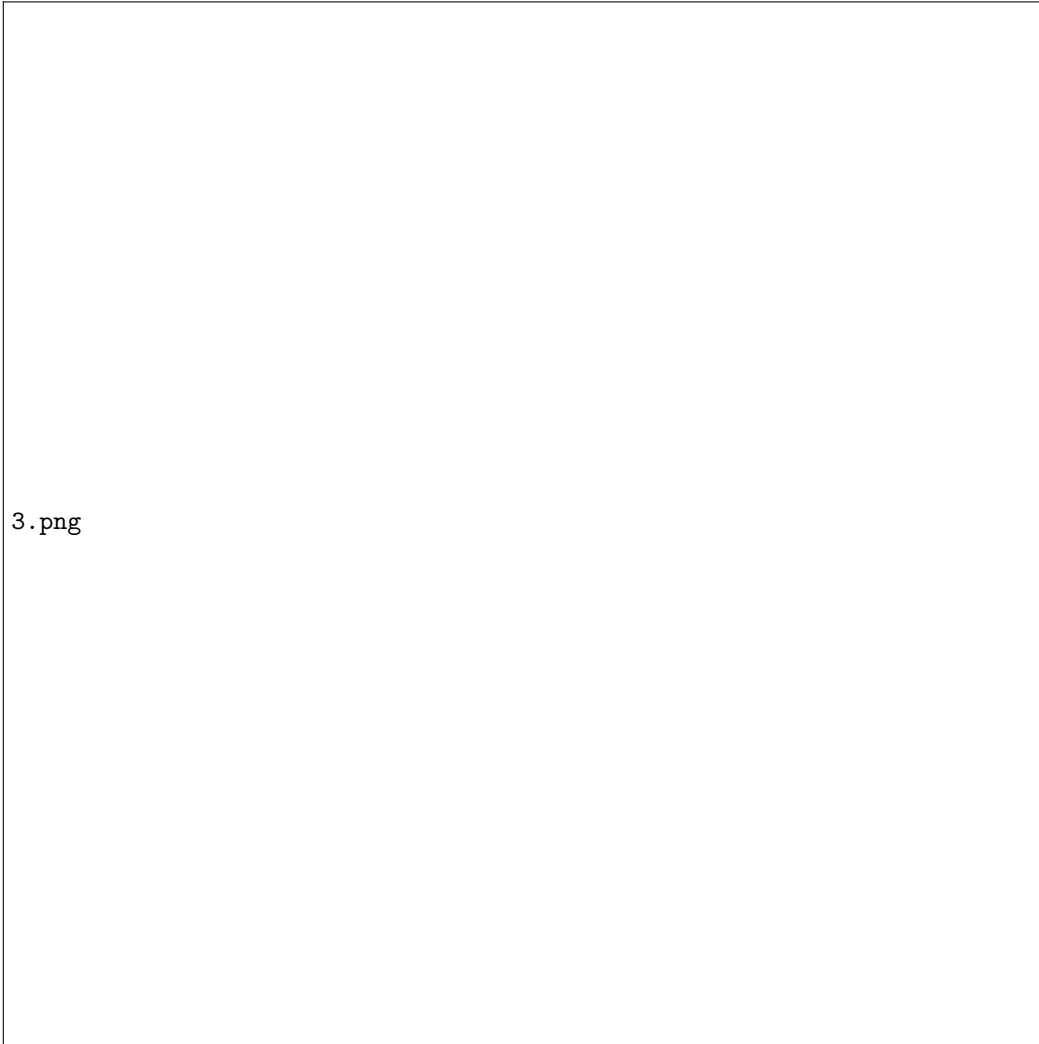


Figure 3:

$$T_F \approx \frac{\epsilon_F}{K} \quad (37)$$

Statistics of Paramagnetism

Paul Haskins

March 27, 2025

N magnetic dipoles interacting with a magnetic $\vec{H}$. Magnetic moment of dipoles $\vec{\mu}$.
We consider in the simplest case that dipoles are fixed and localised.

Completely localised dipoles. No translation only rotation.

Classical treatment

$$E = \sum_{i=1}^N E_i = - \sum_{i=1}^N \vec{\mu} \vec{H} = - \sum_{i=1}^N \mu H \cos \theta_i \quad (1)$$

as all dipoles have the same $|\vec{\mu}| = \mu$.

Thermodynamic limit

$$Z = \prod_{i=1}^N \zeta_i = \zeta^N \quad (2)$$

because there is no interaction between dipoles.

We have the canonical partition function for one dipole

$$\zeta = \zeta(T, V, 1) = \sum_{\theta, \phi} e^{+\beta \vec{\mu} H \cos \theta} = \int d\theta d\phi \sin \theta e^{-\beta \vec{\mu} H \cos \theta} \quad (3)$$

$$= \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta e^{+\beta \vec{\mu} H \cos \theta} \quad (4)$$

$$= (2\pi) \int_{-1}^1 d(\cos \theta) e^{+\beta \vec{\mu} H \cos \theta} \quad (5)$$

if we let $x = \beta \vec{\mu} H \cos \theta$

$$= \frac{2\pi}{\beta \vec{\mu} H} \int_{-\beta \vec{\mu} H}^{\beta \vec{\mu} H} dx e^{-x} = \dots \quad (6)$$

$$\implies \zeta(\beta, V, N=1) = \frac{4\pi \sinh(\beta \vec{\mu} H)}{\beta \vec{\mu} H} \quad (7)$$

$$Z = \zeta^N = \left(\frac{4\pi \sinh(\beta \vec{\mu} H)}{\beta \vec{\mu} H} \right)^N \quad (8)$$

$$\mathcal{F} = -kT \ln Z = -kTN \ln \zeta \quad (9)$$

We can determine the thermo behavior of the system.

$\langle \vec{\mu} \cos \theta \rangle ==$ average dipole moment of each dipole.

$N \langle \vec{\mu} \cos \theta \rangle == \dots$ of the total system.

We can find everything from our free energy in the classical limit.

$$\langle \vec{\mu} \cos \theta \rangle = \frac{\int \int d\phi d\theta \sin \theta \vec{\mu} \cos \theta e^{-\beta \vec{\mu} H \cos \theta}}{\int \int d\phi d\theta \sin \theta e^{-\beta \vec{\mu} H \cos \theta}} \quad (10)$$

$$= \frac{1}{\zeta \beta} \left(\frac{\partial}{\partial H} \zeta \right)_{\beta, V} = \frac{1}{\beta} \left(\frac{\partial \ln \zeta}{\partial H} \right)_{\beta, V} \quad (11)$$

$$\implies -\frac{1}{N} \left(\frac{\partial \mathcal{F}}{\partial H} \right)_{\beta, V, N} == \langle M \rangle \quad (12)$$

χ_t is the isothermal magnetic susceptibility $\chi_T = \left(\frac{\partial \langle M \rangle}{\partial H} \right)_{\beta, V, N}$ We can evaluate

$$\langle M \rangle = \dots = -\frac{1}{\beta H} + \vec{\mu} \coth(\beta \vec{\mu} H) = \vec{\mu} \left(-\frac{1}{\beta \mu H} + \coth(\beta \vec{\mu} H) \right) \quad (13)$$

We can let $x = \beta \vec{\mu} H$ and define the function

$$\mathcal{L}(x) = \left(-\frac{1}{\beta \mu H} + \coth(\beta \vec{\mu} H) \right) \quad (14)$$

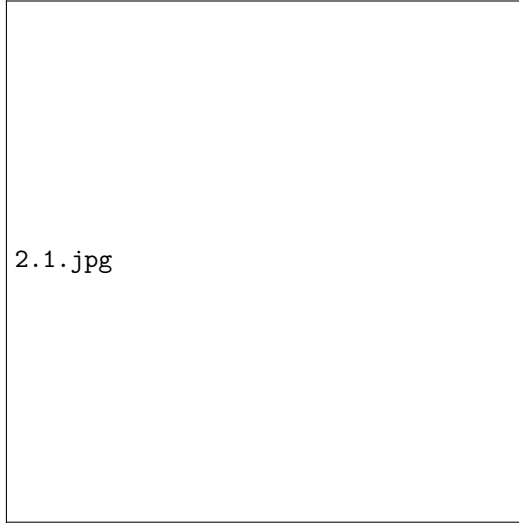


Figure 1: pg 84 beale

$$\hat{H} = -\vec{\mu} \vec{H} \quad (15)$$

quantisation

$$\vec{\mu} = g \left(\frac{q}{2mc} \right) () \quad (16)$$

where g is the gyro magnetic ratio.
either $\vec{L}$, $\vec{S}$ or combination $\vec{J}$.

$$E = -g \vec{\mu}_0 m H \quad (17)$$

m is the quantum number $m = -J, -J+1, \dots, J-1, J$. Notice there are $2J+1$ allowed values.

$$\vec{\mu}_0 \frac{q \hbar}{2mc} \quad (18)$$

$$\zeta(\beta, V, H, N=1) = \sum_{m=-J}^J e^{(\beta g \vec{\mu}_0 H) m} \quad (19)$$

let $x = \beta g \vec{\mu}_0 H J$

$$\zeta = e^{-x} \left(\frac{e^{\frac{2J+1}{J} x} - 1}{e^{\frac{x}{J}} - 1} \right) \quad (20)$$

$$\zeta = \dots = \frac{\sinh \frac{2J+1}{J} x}{\sin \frac{x}{2J}} \quad (21)$$

$$Z = \zeta^N, \mathcal{F} = -kT \ln Z \quad (22)$$

$$\langle M \rangle = - \left(\frac{\partial \ln \zeta}{\partial H} \right)_{\beta, V} \quad (23)$$

$$\chi_T = \left(\frac{\partial \langle M \rangle}{\partial H} \right)_{\beta, V} \quad (24)$$

Localised Dipoles (QM)

Paul Haskins

March 14, 2025

$$\epsilon_m = -g\vec{\mu}mH \quad (1)$$

m is qm number, H is constant magnetic field g is gyro magnetic ratio.

Fully localised, no interaction between particles

$$Z = \zeta^N \zeta(\beta, V, H, N = 1) \quad (2)$$

$$x = \beta g \vec{\mu}_0 J H$$

$$\zeta = \frac{\sinh\left(\frac{\frac{2J+1}{H}X}{\sinh\left(\frac{x}{2J}\right)}\right)}{\sinh\left(\frac{x}{2J}\right)} \quad (3)$$

$$\langle M \rangle = \frac{1}{\beta} \left(\frac{\partial}{\partial H} \ln \zeta \right)_{\beta, V} = g \vec{\mu} J \left[\frac{2J+1}{2J} \coth\left(x \left(\frac{2J+1}{2J} \right)\right) - \frac{1}{2J} \coth\left(\frac{x}{2J}\right) \right] \quad (4)$$

We define the brillion function of order J

$$B_J(x) = \left[\frac{2J+1}{2J} \coth\left(x \left(\frac{2J+1}{2J} \right)\right) - \frac{1}{2J} \coth\left(\frac{x}{2J}\right) \right] \quad (5)$$

We can take limits of $x \gg 1$ Strong magnetic vs temperature.

$$B_J(x) \approx 1, \forall J \quad (6)$$

Magnetic saturation

$x \ll 1$ weak magnetic vs temperature.

$$B_J(x) \approx \frac{1}{3} \left(\frac{J+1}{J} \right) x + \mathcal{O}(x^2) \quad (7)$$

$\Rightarrow$ keeping only the leading term

$$\langle M \rangle \approx \frac{g^2 \vec{\mu}_0^2}{3k} J(J+1) H \frac{1}{T} \quad (8)$$

No qualitative distance but only the pre-factor of $\frac{1}{T}$ different systems have different constants.
This is the difference between the QM and the classical Curie law.

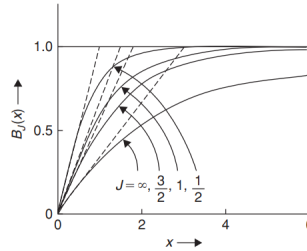


FIGURE 3.8 The Brillouin function $B_J(x)$ for various values of J .

Figure 1: pg 75 beal

When is the classical limit obtained. When we make the spectrum almost continuous. This occurs with a large spectrum and the spectrum is very dense.

We need the following limits $J \rightarrow \infty$ and $g \rightarrow 0$ ($\epsilon_m = g m \vec{\mu}_0 H$).

$B_J(x)$ independent of J .

$$\coth\left(\frac{x}{2J}\right) \approx \frac{2J}{x} \quad (9)$$

with $\frac{x}{2J} \ll 1$
In this limit we have that

$$\langle M \rangle \approx \vec{\mu} \mathcal{L}(x) \quad (10)$$

$$gJ\vec{\mu}_0 == \vec{\mu} \quad (11)$$

classical dipole moment.

There is always a lower bound in E, but now we have an upper bound. This can lead to unusual behavior.

Fix $J = \frac{1}{2}$. $\implies$ only 2 values for the energy. (fermions)

$$\epsilon_i = \begin{cases} -\vec{\mu}_0 H & == -\epsilon_0 \\ +\vec{\mu}_0 H & == \epsilon_0 \end{cases} \quad (12)$$

Thermodynamic limit - canonical ensemble

$$Z = \zeta^N = \left(\frac{\sinh 2x}{\sinh x} \right)^N = 2^N \cosh^N(x), x == \beta \vec{\mu}_0 H = \beta \epsilon_0 \quad (13)$$

$$\mathcal{F} = -kT \ln Z = -kTN \ln [2 \cosh \beta \epsilon_0] \quad (14)$$

$$S = - \left(\frac{\partial \mathcal{F}}{\partial T} \right)_{V,H,N} = NK [\ln (2 \cosh (\beta \epsilon_0)) - \beta \epsilon_0 \tanh (\beta \epsilon_0)] \quad (15)$$

$$\langle E \rangle = F + TS = - \frac{\partial \ln Z}{\partial \beta} = -N \epsilon_0 \tanh (\beta \epsilon_0) = -N \langle M \rangle H \quad (16)$$

$$\langle M \rangle = - \frac{1}{N} \left(\frac{\partial \mathcal{F}}{\partial H} \right)_{\beta,V,N} = \vec{\mu}_0 \tanh (\beta_0) \quad (17)$$

$$C_V = \left(\frac{\partial \langle E \rangle}{\partial T} \right)_{V,H,N} = \frac{Nk (\beta \epsilon_0)^2}{\cosh^2 (\beta \epsilon_0)} \quad (18)$$

thus in the limit $\beta \epsilon_0 \ll 1$

$$S \approx Nk [\ln (2) - (\beta \epsilon_0)^2] \approx Nk \ln 2 \quad (19)$$

and if $\beta_0 \gg 1$

$$S \approx Nk [\beta \epsilon_0 - \beta \epsilon_0] = 0 \quad (20)$$

doesn't only happens for low temp but also for strong magnetic fields.

We now examine in the micro canonical ensemble

$$S = k \ln \Omega \quad (21)$$

for the second S there would only be one microstate. We can also examine for the first S.

$$\Omega = \begin{cases} 2^N & \text{if } kT \gg \epsilon_0 \\ 1 & kT \ll \epsilon_0 \end{cases}$$

(23)

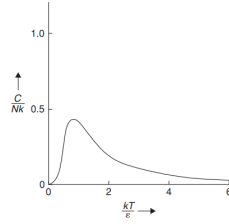


FIGURE 3.13 The specific heat of a system of magnetic dipoles (with $J = \frac{1}{2}$) as a function of temperature.

Figure 2: 3.2.png

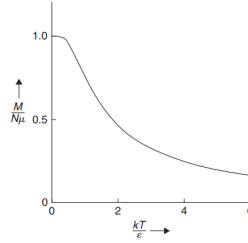


FIGURE 3.12 The magnetization of a system of magnetic dipoles (with $J = \frac{1}{2}$) as a function of temperature.

Figure 3: 3.2.png

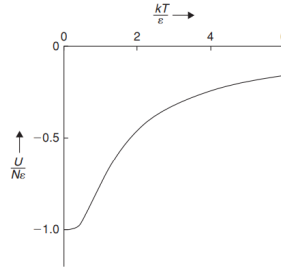


Figure 4: 3.2.png

Solve for $T \langle E \rangle \Rightarrow$

$$\frac{1}{T} = \frac{k}{2\epsilon_0} \ln \left[\frac{1 - \frac{\langle E \rangle}{N\epsilon_0}}{1 + \frac{\langle E \rangle}{N\epsilon_0}} \right] \quad (24)$$

* * * $T < 0$! Characteristic feature of systems who energy has an upper bound .

$$S = -\frac{N\epsilon_0 \langle E \rangle}{2N\epsilon_0} \ln \left[\frac{N\epsilon_0 + \langle E \rangle}{2N\epsilon_0} \right] - \frac{N\epsilon_0 - \langle E \rangle}{2N\epsilon_0} \ln \left[\frac{N\epsilon_0 - \langle E \rangle}{2N\epsilon_0} \right] \quad (25)$$

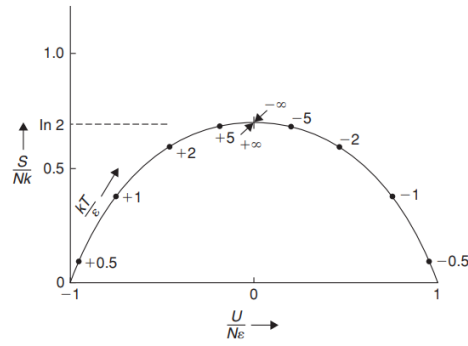


FIGURE 3.14 The entropy of a system of magnetic dipoles (with $f = \frac{1}{2}$) as a function of energy. Some values of the parameter kT/ϵ are also shown in the figure. The slope at the two endpoints diverges since both ends represent zero temperature but it is difficult to see due to the logarithmic nature of the divergence.

Figure 5: the slope of the graph is $\frac{1}{T}$. $\langle E \rangle = -N\epsilon_0$, $S = 0$, $T = 0$.
 as $\langle E \rangle$ increases, S and T increase. Max is reached as $T \rightarrow \infty$. $s \approx Nk \ln 2$

But when $\langle E \rangle \rightarrow 0^+$, then $\frac{\partial S}{\partial E} \propto \frac{1}{T} < 0$.
 Further increasing $\langle E \rangle$ $S(E)$ decreases and T remains negative.

The RHS has hotter systems. T increase on left up to infity on left.
 If we add a kinetic term to hamitonian then we no longer have an upper bound.

Systems whos energy has an upper bound

Paul Haskins

March 20, 2025

The fact that there is an upper bound is related to the appearance of negative temperature.
In this case $\exists T$ such that

$$kT \gg \frac{E}{N} \quad (1)$$

Consider canonical partition function

$$Z \approx \left(\sum_n^e -\beta \epsilon_n \right)^N \quad (2)$$

(neglecting interactions for high T) $\beta \epsilon_n \ll 1$

$$\Rightarrow Z \approx \left(\sum_n \left(1 - \beta \epsilon_n + \frac{1}{2} \beta^2 \epsilon_n^2 \right) \right)^N \quad (3)$$

$$\ln Z \approx N \ln \left[\sum_n 1 - \sum_n \beta \epsilon_n + \frac{1}{2} \sum_n \beta^2 \epsilon_n^2 \right] \quad (4)$$

To proceed we set $\sum_n 1 \approx g$ e.g. all possible orientations in a magnetic field, all possible polarization's etc.

$$\sum_n \beta \epsilon_n \approx g \beta \langle \epsilon_n \rangle \quad (5)$$

$$\sum_n \beta^2 \epsilon_n^2 \approx g \beta^2 \langle \epsilon_n^2 \rangle \quad (6)$$

Subbing into $\ln Z$.

$$\ln Z \approx N \ln \left[g \left(1 - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 \langle \epsilon_n^2 \rangle \right) \right] \quad (7)$$

$$\approx N \left[\ln g - \left(\beta \langle \epsilon_n \rangle - \frac{1}{2} \beta^2 \langle \epsilon_n^2 \rangle \right) - \frac{1}{2} \left(\beta \langle \epsilon_n \rangle - \frac{1}{2} \beta^2 \langle \epsilon_n^2 \rangle \right)^2 + \dots \right] \quad (8)$$

$$\approx N \left[\ln g - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 (\langle \epsilon_n^2 \rangle - \langle \epsilon_n \rangle^2) \right] \quad (9)$$

so we can use

$$\ln Z \approx N \left[\ln g - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 (\langle \epsilon_n^2 \rangle - \langle \epsilon_n \rangle^2) \right] \quad (10)$$

into

$$\mathcal{F} = -kT \ln Z \quad (11)$$

$$\approx -NkT \ln g + N \langle \epsilon_n \rangle - \frac{N}{2kT} \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad (12)$$

$$S \approx Nk \ln g - \frac{Nk}{2} \beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad (13)$$

$$\langle E \rangle \approx N \langle \epsilon_n \rangle - N \beta \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad (14)$$

$$C_V \approx Nk \beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \geq 0 \quad (15)$$

These formulae determine the thermodynamic behavior of the system, for $\beta \rightarrow 0^+$ or $\beta \rightarrow 0^-$

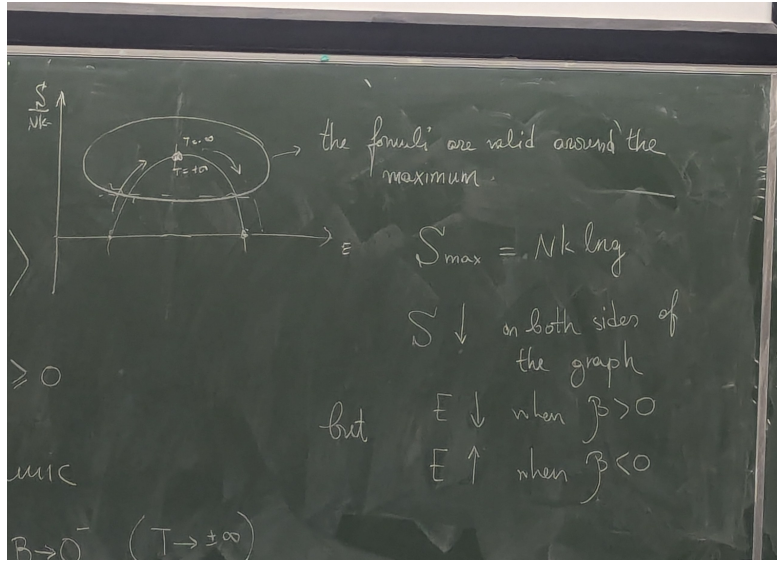


Figure 1: 4.1.png

Exercise

Ideal gas of (indistinguishable) electrons in a constant magnetic field $\vec{H}$ enclosed in a volume V .

Compute the average dipole moment and the magnetic susceptibility weak or strong magnetic fields, large or small T , Curie Law?.

$$\ln \Xi = \sum_n \ln (1 + e^{-\beta \epsilon_n} e^{\beta \mu}) \quad (16)$$

+ in log since we have fermions
We add a magnetic field so

$$e^{-\beta \epsilon_n} \rightarrow e^{-\beta \epsilon_n(p) \pm \beta \vec{\mu} \cdot \vec{H}} \quad (17)$$

$$\ln \Xi = \sum_n \ln (1 + e^{-\beta \epsilon_n(p) \pm \beta \vec{\mu} \cdot \vec{H}} e^{\beta \mu}) \quad (18)$$

$$= \sum_n \ln (1 + e^{-\beta \epsilon_n(p)} e^{\beta \mu_+}) + \sum_n \ln (1 + e^{-\beta \epsilon_n(p)} e^{\beta \mu_-}) \quad (19)$$

$$= \sum_n \ln (1 + z_+ e^{-\beta \epsilon_n(p)}) + \sum_n \ln (1 + z_- e^{-\beta \epsilon_n(p)}) \quad (20)$$

The net effect of introducing a magnetic field to shift a chemical potential μ Where we have defined

$$\mu_{\pm} = \mu \pm \vec{\mu} \cdot \vec{H} \quad (21)$$

$$z_{\pm} = e^{\beta \mu_{\pm}} = e^{\beta (\mu \pm \vec{\mu} \cdot \vec{H})} \quad (22)$$

Where we have the $+$ $\implies$ disalignment and $-$ $\implies$ alignment

$$J = -kT \ln \Xi \approx (\text{deg}) \frac{kTV}{\lambda_T^3} \left(\mathcal{L}i_{\frac{5}{2}}(-z_+) + \mathcal{L}i_{\frac{5}{2}}(-z_-) \right) \quad (23)$$

$$\lambda_T = \left(\frac{h^2}{2\pi m kT} \right) \quad (24)$$

$$\langle M \rangle = -\frac{1}{N} \left(\frac{\partial J}{\partial H} \right)_{T, \mu, V} \quad (25)$$

$$\chi_T = \left(\frac{\partial \langle M \rangle}{\partial H} \right)_{T, V, N} \quad (26)$$

$$\langle M \rangle = \dots = \frac{\vec{\mu}}{N} (N_+ - N_-) \quad (27)$$

we can associate $\langle N_+ \rangle$ with the poly log of $-z_+$ and similarly for $-z_-$.

For small temperatures

$$\chi_T \approx -\frac{V}{N} \frac{\vec{\mu}}{\lambda_T^3} (\beta \vec{\mu}) \left[\mathcal{L}i_{\frac{1}{2}}(-z_+) + \mathcal{L}i_{\frac{1}{2}}(-z_-) \right] \approx -\frac{2\beta \vec{\mu}^2 V}{\lambda_T^3} \mathcal{L}i_{\frac{1}{2}}(-z) \quad (28)$$

Then we can show that the curie law is valid.

If we had bosons we would have 3 possible orientations 3 possible terms to add to the energy 3 polylogs. Would we have bose einstein condensation. for which orientation would be occur.

Ising model 1D

Paul Haskins

March 21, 2025

Discrete model of spins ± 1 localised at specific points on a lattice.

$$H = - \sum_{ij}^N J_{ij} s_i s_j - \vec{\mu} \sum_i^N H_i s_i + \sum_{ijk}^N K_{ijk} s_i s_j s_k + \dots \quad (1)$$

$$s_i = \pm 1 \quad (2)$$

The first term is a new 'pairwise' interaction We call these all terms coupling constants.

$$J_{ij} = \begin{cases} > 0 \forall i, j & \text{if ferramagnetic} \\ < 0 \forall i, j & \text{anti-ferromagnetic} \\ g(x) & \text{otherwise} \\ 0, \text{non-interacting} \end{cases} \quad (3)$$

For the thermodynamic limit to exist for this system this condition must hold for at lease the pairwise coupling $\sum_{ij} |J_{ij}| < \infty$

We will typically study in canonical ensemble. For thermodynamic limit $N \gg 1$. What does it mean for the limit to exist? May not necessarily exist, previously we took the leading term but here when we take a limit it may not exist. We may have to impose a constraint like the one above.

We will make the simplification for now where we only study the nearest neighbor interactions. $J_{ij} = J > 0$.

$$H = -J \sum_{i=1} s_i s_{i+1} - \vec{\mu} \sum_{i=1} H_i s_i \quad (4)$$

Canonical Partition function (for diff B.C.)

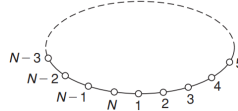


FIGURE 13.3 An Ising chain with a closed, endless structure.

Figure 1: 5.1.png

For higher Dimensions we can think of surface issues as B.C.

Now we will compute the free boundary condition H_i (an infinite chain).

$$\mathcal{Z} = \sum_{\{s_i\}} e^{\beta J \sum_{i=1}^{N-1} s_i s_{i+1}} \quad (5)$$

Note only up to $N - 1$ as the $i + 1$ term would account for the final spin.

$$\mathcal{Z} = \sum_{s_1} \dots \sum_{s_n} e^{\beta J (s_1 s_2 + s_2 s_3 + \dots + s_{N-1} s_N)} \quad (6)$$

Call $\eta_i = s_i s_{i+1} = \begin{cases} +1 & \text{if } s_i = s_{i+1} \\ -1 & \text{if } s_i \neq s_{i+1} \end{cases}$ We replace the $N-1$ sums by sums over η_i and keep the sum over s_i .

$$\Rightarrow \mathcal{Z} = \sum_{s_1} \sum_{\eta_1} \dots \sum_{\eta_{N-1}} e^{\beta J (\eta_1 + \eta_2 + \dots + \eta_{N-1})} \quad (7)$$

$$= \sum_{s_1 = \pm 1} \left[\prod_{i=1}^{N-1} \left(\sum_{\eta_i = \pm 1} e^{\beta J \eta_i} \right) \right] \quad (8)$$

$$= \sum_{s_1 = \pm 1} \left(\sum_{\eta = \pm 1} e^{\beta J \eta} \right)^{N-1} \quad (9)$$

$$= \sum_{s_1 = \pm 1} (2 \cosh(\beta J))^{N-1} \quad (10)$$

$$\Rightarrow \mathcal{Z} = 2 (2 \cosh(\beta J))^{N-1} \quad (11)$$

$H_i = 0$ free B.C. $\mathcal{Z} = 2 (2 \cosh(\beta J))^{N-1}$ periodic B.C.

Periodic B.C. we want last particle to interact with the first one

$$\mathcal{Z} = \sum_{\{s_i\}} e^{\beta J \sum_{i=1}^{N-1} s_i s_{i+1} + \beta J s_N s_1} \quad (12)$$

Suppose we introduce $\eta_i = s_i s_{i+1}$ How do we express $s_1 s_N$ in terms of η_i ?
we know the square must always be 1. so we can introduce these 1's.

$$s_N s_1 = s_1 s_2 s_2 s_3 s_3 \dots s_{N-1} s_{N-1} s_N = \eta_1 \dots \eta_{N-1} \quad (13)$$

Let us employ this change of variables $\{s_i\} \rightarrow \{\eta_i, s_1\}$.

$$\mathcal{Z} = \sum_{s_1} \sum_{\eta_1} \dots \sum_{\eta_{N-1}} e^{\beta J \eta_1} \dots e^{\beta J \eta_{N-1}} e^{\beta J \eta_1 \eta_2 \dots \eta_{N-1}} \quad (14)$$

$$= \sum_{s_1} \sum_{\eta_1} \dots \sum_{\eta_{N-1}} e^{\beta J \eta_1} e^{\beta J \eta_2} \dots e^{\beta J \eta_{N-1}} \cdot \sum_{\alpha=0}^{\infty} \frac{(\beta J)^\alpha}{\alpha!} (\eta_1 \eta_2 \dots \eta_{N-1})^\alpha \quad (15)$$

Exchanging the order of summation

$$\mathcal{Z} = \sum_{s_1} \sum_{\alpha=0}^{\infty} \frac{(\beta J)^\alpha}{\alpha!} \left(\sum_{\eta = \pm 1} e^{\beta J \eta} \eta^\alpha \right)^{N-1} \quad (16)$$

$$= \sum_{s_1 = \pm 1} \sum_{\alpha=0}^{\infty} \frac{(\beta J)^\alpha}{\alpha!} \left(e^{\beta J} + (-1)^\alpha e^{-\beta J} \right)^{N-1} \quad (17)$$

Note that

$$e^{\beta J} + (-1)^\alpha e^{-\beta J} = \begin{cases} \alpha = \text{even} & \text{if } 2 \cosh(\beta J) \\ \alpha = \text{odd} & 2 \sinh(\beta J) \end{cases} \quad (18)$$

$$\mathcal{Z} = \sum_{s_1 = \pm 1} \left[\sum_{n=0}^{\infty} \frac{(\beta J)^{2n}}{(2n)!} (2 \cosh(\beta J))^{N-1} + \sum_{n=0}^{\infty} \frac{(\beta J)^{2n+1}}{(2n+1)!} (2 \sinh(\beta J))^{N-1} \right] \quad (19)$$

$$= \sum_{s_1=\pm 1} [2^{N-1} \cdot \cosh^N(\beta J) + 2^{N-1} \sinh^N(\beta J)] \quad (20)$$

$$\Rightarrow \mathcal{Z} = 2^N \cosh^N(\beta J) + 2^N \sinh^N(\beta J) \quad (21)$$

Thus we have for periodic B.C.

$$\mathcal{Z} = (2 \cosh(\beta J))^N + (2 \sinh(\beta J))^N \quad (22)$$

We have the Helmholtz

$$\mathcal{F}(g_i, \beta, V, N) \quad (23)$$

where g_i is the coupling of the system.

We want to confirm the log of z behaves the same in the thermodynamic limit.

(a)

Free B.C.

$$\mathcal{F} = -kT \ln Z = -kT [\ln 2 + (N-1) \ln(2 \cosh(\beta J))] \quad (24)$$

$$= -kTN \left[\frac{1}{N} \ln 2 + \frac{N-1}{N} \ln(2 \cosh(\beta J)) \right] \rightarrow \mathcal{F}_A \approx -kTN \ln(2 \cosh(\beta J)) \quad (25)$$

for $N \gg 1$

(b)

Periodic B.C.

$$\mathcal{F} = -kT \ln [2^N \cosh^N(\beta J) [1 + \tanh^N(\beta J)]] \quad (26)$$

$$= -kTN \ln(2 \cosh(\beta J)) - kT \ln [1 + \tanh^N(\beta J)] \quad (27)$$

$$\mathcal{F}_B \approx -kTN \ln(2 \cosh(\beta J)) \quad (28)$$

for $N \gg 1$.

Where we have used

$$\ln [1 + \tanh^N(\beta J)] \approx \dots \quad (29)$$

$$0 < |\tanh(\beta J)| < 1 \quad (30)$$

So if we are in the thermodynamic limit we don't have to stick to one type of boundary conditions, we can pick the one which makes it easier for computations.

(c)

$H_i = H \neq 0$ Chose to study with periodic B.C.

Method: transfer matrix.

$$\mathcal{Z} = \sum_{s_1} \dots \sum_{s_N} e^{\beta H \sum_{i=1}^N s_i + \beta J \sum_{i=1}^N s_i s_{i+1}} \quad (31)$$

$$= \sum_{s_1} \dots \sum_{s_N} e^{\frac{1}{2}\beta H(s_1+s_2)+\beta J s_1 s_2} \dots e^{\frac{1}{2}\beta H(s_N+s_1)+\beta J s_N s_1} \quad (32)$$

We will identify each of these terms with a component of a matrix element.

$$e^{\frac{1}{2}\beta H(s_1+s_2)+\beta J s_1 s_2} \equiv \mathcal{T}_{(s_1 s_2)} : \begin{bmatrix} e^{\beta H + \beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{-\beta H + \beta J} \end{bmatrix} \quad (33)$$

Substituting

$$\mathcal{Z} = \sum_{s_1} \sum_{s_N} \mathcal{T}_{(s_1 s_2)} \mathcal{T}_{(s_2 s_3)} \cdots \mathcal{T}_{(s_{N-1} s_N)} \mathcal{T}_{(s_N s_1)} \quad (34)$$

If we consider all the sums for all the terms we get a product

$$= \sum_{s_1} \left(\mathcal{T}^N \right)_{(s_1 s_1)} \quad (35)$$

where the sum is just the trace

$$\implies \mathcal{Z} = \text{Tr} [\mathcal{T}^N] \quad (36)$$

$$\mathcal{T} = \begin{bmatrix} e^{\beta(H+J)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(J-H)} \end{bmatrix} \quad (37)$$

$\mathcal{T}$ is real and symmetric $\rightarrow$ diagonalisable by some $\mathcal{O}$ with $\mathcal{O} = \mathcal{O}^T$. Then if

$$\mathcal{T}_{\text{diagonal}} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \quad (38)$$

$$\implies \mathcal{Z} = \text{Tr} [\mathcal{T}^N] = \lambda_1^N + \lambda_2^N \quad (39)$$

There are two cases either $\lambda_1 = \lambda_2$ or $\lambda_1 \neq \lambda_2$. where second case actually never happens. then if $\lambda_1 > \lambda_2$

$$\implies \mathcal{Z} = \lambda_1^N \left[1 + \left(\frac{\lambda_2}{\lambda_1} \right)^N \right] \quad (40)$$

$$\ln \mathcal{Z} = N \ln \lambda_1 + \ln \left(1 + \left(\frac{\lambda_2}{\lambda_1} \right)^N \right) \quad (41)$$

where the second term is bounded.

In the thermodynamic limit only **the largest** λ_i matters.

$$\lambda_{1,2} = e^{\beta J} \left[\cosh(\beta H) \pm \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right] \quad (42)$$

$$\implies \mathcal{F} = -NJ - NkT \ln \left[\cosh(\beta J) + \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right] \quad (43)$$

The first term is the energy, the log with sqrt means there may be non analytic $\mathcal{F}$ which leads to the potential for phase transitions.

In general if in the thermodynamic limit the thermodynamic potential exists (in our case Helmholtz) $\mathcal{F}(g_i)$, $i = 1, \dots, D$.

Phase diagram with D-axes.

Possible non-analyticities can be points, lines, planes etc.

Phase boundaries correspond necessarily to codimension-1 non-analytic regions. Codimensions is the total dimension of the system minus the dimension of the object.

The only non-analyticities are with the derivative not with $\mathcal{F}$ itself.

1D ising model

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$$H = - \sum_{ij} J_{ij} s_i s_j - \tilde{\mu} \sum_i H_i s_i + \mathcal{O}(s_i s_j s_k) \quad (1)$$

We discussed the homogeneous case

$$J_{ij} = \begin{cases} J & \text{if } \forall i \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

And $h_i \equiv H$ constant magnetic field.

Generally we have the following conditions

→ Boundary conditions do not matter in the thermodynamic limit.

→ Transfer matrix $\mathcal{T} = \begin{bmatrix} e^{\beta H + \beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{-\beta H + \beta J} \end{bmatrix}$ We showed $Z = \text{tr}(\mathcal{T}^N) = \lambda_1^N + \lambda_2^N$ where $\lambda_{1,2}$ are eigenvalues of $\mathcal{T}$.

$$\lambda_{1,2} = e^{\beta J} \left(\cosh(\beta H) \pm \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right) \quad (3)$$

In the thermodynamic limit assuming $\lambda_1 > \lambda_2$

$$\implies \mathcal{F} = -kTN \ln \lambda_1 \quad (4)$$

Assume that thermodynamic potential e.g $\mathcal{F}$ exists in the thermodynamic limit,

$$\mathcal{F}(g_i), \{g_i\} \equiv \text{couplings} \quad (5)$$

$\mathcal{F}$ is analytic except at some points, lines, hyper surfaces in the space of couplings $\{g_i\}$.

Then we can define the 'phase boundary' as a co-dimension one surface in coupling space where $\mathcal{F}$ is not analytic (non analytic surface is the dimension of the space minus one).

It can be shown that $\mathcal{F}$ is continuous

$$\frac{\partial \mathcal{F}}{\partial g^i} \text{ discontinuous} \implies \text{1st order p.t} \quad (6)$$

$$\frac{\partial \mathcal{F}}{\partial g^i} \text{ not discontinuous} \implies \text{continuous p.t.} \quad (7)$$

Properties of $\mathcal{F}$ in 1D- ising

$$\mathcal{F} < 0, \text{continuous} \quad (8)$$

$$\frac{\partial \mathcal{F}}{\partial g^i} \text{ exists almost everywhere} \quad (9)$$

Left-right derivatives exist and are equivalent everywhere

$$S = -\frac{\partial \mathcal{F}}{\partial T} \geq 0 \quad (10)$$

$$\frac{\partial^2 \mathcal{F}}{\partial T^2} \leftrightarrow C_V \geq 0 \quad (11)$$

$$\frac{\partial^2 \mathcal{F}}{\partial H^2} \leq 0 \leftrightarrow \chi_T \geq 0 \quad (12)$$

Are these transitions possible i.e., are there regions of non-analytic city of $\mathcal{F}$?

Recall

$$\mathcal{F} = -kTN \ln \lambda_1, \lambda_1 > \lambda_2 \quad (13)$$

$\mathcal{F}$ might be non-analytic iff

λ_1 is not analytic

$\lambda_1 = \lambda_2$ or

$\lambda_1 = 0$

Perron's theorem

A nxn matrix $\mathcal{A}$ with $\mathcal{A}_{i,j} > 0 \forall i, j$ then $\exists \lambda_1 > \forall j$ such that

$\lambda_1 > 0$

non- degenerate ($\lambda_1 \neq \lambda_2$)

λ_1 is analytic function of $\mathcal{A}_{ij}$

These criteria rule out the possibility for non-analytic behavior of $\mathcal{F} \implies$ no phase transition. This is true for $T > 0$.

We are interested in the regime where the theorem may not be applicable in $T = 0$ or $\beta \rightarrow \infty$ or $J \rightarrow \infty, \beta J \gg 1$

$$\lambda_1 = e^{\beta J} \left(\cosh(\beta H) + \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right) \quad (14)$$

for $\beta J \gg 1$

$$\approx e^{\beta J} (\cosh(\beta H) + \sinh(\beta H)) \quad (15)$$

$$\rightarrow \lambda_1 \approx e^{\beta J + |\beta H|} \quad (16)$$

$$\mathcal{F} = -kNT \ln \lambda_1 \approx -N(J + |H|) \quad (17)$$

$$\langle M \rangle = -\frac{1}{N} \frac{\partial \mathcal{F}}{\partial H} = \begin{cases} 1 & \text{if } h > 0 \\ -1 & \text{if } h < 0 \end{cases} \quad (18)$$

Discontinuity in the derivative of $\mathcal{F}$! $\implies$ p.t at $T = 0$!

we want to consider correlations between our thermodynamic quantities.

$$\langle s_i s_j \rangle \leftrightarrow \text{discrete analog of two-point correlators in QFT (or propagators)} \quad (19)$$

j here isn't necessarily $i \pm 1$ rather it can be arbitrary.

We define the correlation functions or two-point functions

$$G_{ij} \equiv G(i, j) = \langle s_i s_j \rangle - \langle s_i \rangle \langle s_j \rangle \quad (20)$$

$$= \langle (s_i - \langle s_i \rangle) (s_j - \langle s_j \rangle) \rangle \quad (21)$$

$$= \langle s_i s_j \rangle \quad (22)$$

because here $\langle s_i \rangle = 0 \forall i = 1, 2, \dots, N$

$G_{i,j}$ is also related to the probability $Pr(s_i = s_j)$

$$Pr_{ij} = \frac{1}{Z} \sum_{ij} \delta_{s_i, s_j} e^{-\beta H} = \langle \delta_{s_i s_j} \rangle = \left\langle \frac{1}{2} (1 + s_i s_j) \right\rangle \quad (23)$$

$$\implies Pr_{ij} = \frac{1}{2} + \frac{1}{2} G_{ij} \quad (24)$$

How to compute G_{ij}

Set $H \neq 0$

first compute $G_{i,i+1}$

$$G(i, i+1) = \frac{1}{Z} \sum_{s_1, s_2, \dots, s_N} s_i s_{i+1} e^{\beta J s_1 s_2 + \dots + \beta J s_{N-1} s_N} \quad (25)$$

$$(26)$$

we want to bring down some part so we take derivative

$$= \frac{1}{Z} \frac{\partial}{\partial (\beta J_i)} \left(\sum_{s_1, \dots, s_N} e^{\beta J_1 s_1 s_2 + \dots + \beta J_{N-1} s_{N-1} s_N} \right) \quad (27)$$

$$G(i, i+1) = \frac{\partial}{\partial (\beta J_i)} \ln Z(J_{i,j} \beta) \quad (28)$$